The Victorian
Curriculum

Subjects	Mathematics
Year levels	Level A, Level B, Level C, Level D, Foundation Level, Level 1, Level 2, Level 3, Level 4, Level 5, Level 6, Level 7, Level 8, Level 9, Level 10, Level 10A
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Victorian Curriculum
Mathematics

Victorian Curriculum
Foundation–10

Overview

About Mathematics

The VCAA is developing the Victorian Curriculum F–10 Version 2.0, starting with the publication of the Mathematics Version 2.0 curriculum in Term 3 2023.

You can access both the current Mathematics curriculum and Mathematics Version 2.0 using the menu on this page.

Go to the VCAA website to find information about timelines for Mathematics Version 2.0 and the rest of the Victorian Curriculum F–10 Version 2.0, to register for professional learning webinars and to find more supporting resources.

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Victorian Curriculum
Mathematics

Victorian Curriculum
Foundation–10

Overview

Rationale and Aims

The VCAA is developing the Victorian Curriculum F–10 Version 2.0, starting with the publication of the Mathematics Version 2.0 curriculum in Term 3 2023.

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Rationale

Mathematics provides students with access to important mathematical ideas, knowledge and skills that they will draw on in their personal and work lives. The curriculum also provides students, as life-long learners, with the basis on which further study and research in mathematics and applications in many other fields are built.

Mathematical ideas have evolved across societies and cultures over thousands of years, and are constantly developing. Digital technologies are facilitating this expansion of ideas and provide new tools for mathematical exploration and invention. While the usefulness of mathematics for modelling and problem solving is well known, mathematics also has a fundamental role in both enabling and sustaining cultural, social, economic and technological advances and empowering individuals to become critical citizens.

Number, measurement and geometry, statistics and probability are common aspects of most people's mathematical experience in everyday personal, study and work situations. Equally important are the essential roles that algebra, functions and relations, logic, mathematical structure and working mathematically play in people's understanding of the natural and human worlds, and the interaction between them.

The Mathematics curriculum focuses on developing increasingly sophisticated and refined mathematical understanding, fluency, reasoning, modelling and problem-solving. These capabilities enable students to respond to familiar and unfamiliar situations by employing mathematics to make informed decisions and solve problems efficiently.

The curriculum ensures that the links between the various components of mathematics, as well as the relationship between mathematics and other disciplines, are made clear. Mathematics is composed of multiple but interrelated and interdependent concepts and structures which students apply beyond the mathematics classroom. For example, in Science, understanding sources of error and their impact on the confidence of conclusions is vital; in Geography, interpretation of data underpins the study of human populations and their physical environments; in History, students need to be able to imagine timelines and time frames to reconcile related events; and in English, deriving quantitative, logical and spatial information is an important aspect of making meaning of texts.

Aims

The Mathematics curriculum aims to ensure that students:

- develop useful mathematical and numeracy skills for everyday life, work and as active and critical citizens in a technological world
- see connections and apply mathematical concepts, skills and processes to pose and solve problems in mathematics and in other disciplines and contexts
- acquire specialist knowledge and skills in mathematics that provide for further study in the discipline
- appreciate mathematics as a discipline – its history, ideas, problems and applications, aesthetics and philosophy.

Structure

The curriculum is organised by the three strands of Number and Algebra, Measurement and Geometry, and Statistics and Probability.

Each strand is organised by sub--strands. Sub-strands group content descriptions under an appropriate concept, to provide both a focus and a clear sequence for the development of related concepts and skills within strands and across levels.

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Strands	Number and Algebra	Measurement and Geometry	Statistics and Probability
Sub-strands	Number and place value	Using units of measurement	Chance
	Fractions and decimals	Shape	Data representation and interpretation
	Real numbers	Geometric reasoning	
	Money and financial mathematics	Location and transformation	
	Patterns and algebra	Pythagoras and trigonometry	
	Linear and non--linear relationships		

Number and Algebra

Number and Algebra are developed together, and each enriches the study of the other. Students apply number sense and strategies for counting and representing numbers. They explore the magnitude and properties of numbers. They apply a range of strategies for computation and understand the connections between operations. They recognise patterns and understand the concepts of variable and function. They build on their understanding of the number system to describe relationships and formulate generalisations. They recognise equivalence and solve equations and inequalities. They apply their number and algebra skills to conduct investigations, solve problems and communicate their reasoning.

Measurement and Geometry

Measurement and Geometry are presented together to emphasise their relationship to each other, enhancing their practical relevance. Students develop an increasingly sophisticated understanding of size, shape, relative position and movement of two-dimensional figures in the plane and three-dimensional objects in space. They investigate properties and apply their understanding of them to define, compare and construct figures and objects. They learn to develop geometric arguments. They make meaningful measurements of quantities, choosing appropriate metric units of measurement. They build an understanding of the connections between units and calculate derived measures such as area, speed and density.

Statistics and Probability

Statistics and Probability develops initially in parallel, with the curriculum progressively building links between them. Students recognise and analyse data and draw inferences. They represent, summarise and interpret data and undertake purposeful investigations involving the collection and interpretation of data. Students recognise variation, assess likelihood and assign probabilities using experimental and theoretical approaches. They develop an increasingly sophisticated ability to critically evaluate chance and data concepts and make reasoned judgments and decisions, as well as building skills to critically evaluate statistical information and develop intuitions about data.

Achievement standards

In Mathematics, students progress along a curriculum continuum that provides the first achievement standard at Foundation and then at Levels 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10. Level 10 also has an optional Level 10A, which is intended for students requiring further mathematical studies.

A 'Towards Foundation Levels A to D' curriculum is provided for students with disabilities or additional learning needs in this curriculum area.

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Learning in Mathematics

The proficiencies of Understanding, Fluency, Problem Solving and Reasoning are fundamental to learning mathematics and working mathematically and are applied across all three strands Number and Algebra, Measurement and Geometry, and Statistics and Probability.

Understanding refers to students building a robust knowledge of adaptable and transferable mathematical concepts and structures. Students make connections between related concepts and progressively apply the familiar to develop new ideas. They develop an understanding of the relationship between the 'why' and the 'how' of mathematics. Students build understanding when they:

- connect related ideas
- represent concepts in different ways
- identify commonalities and differences between aspects of content
- describe their thinking mathematically
- interpret mathematical information.

Fluency describes students developing skills in choosing appropriate procedures, carrying out procedures flexibly, accurately, efficiently and appropriately, and recalling factual knowledge and concepts readily. Students are fluent when they:

- make reasonable estimates
- calculate answers efficiently
- recognise robust ways of answering questions
- choose appropriate methods and approximations
- recall definitions and regularly use facts,
- can manipulate expressions and equations to find solutions.

Problem-solving is the ability of students to make choices, interpret, formulate, model and investigate problem situations, select and use technological functions and communicate solutions effectively. Students pose and solve problems when they:

- use mathematics to represent unfamiliar or meaningful situations
- design investigations and plan their approaches
- apply their existing strategies to seek solutions
- verify that their answers are reasonable.

Reasoning refers to students developing an increasingly sophisticated capacity for logical, statistical and probabilistic thinking and actions, such as conjecturing, hypothesising, analysing, proving, evaluating, explaining, inferring, justifying, refuting, abstracting and generalising. Students are reasoning mathematically when they:

- explain their thinking
- deduce and justify strategies used and conclusions reached
- adapt the known to the unknown
- transfer learning from one context to another
- prove that something is true or false
- make inferences about data or the likelihood of events
- compare and contrast related ideas and explain their choices.

Information Communication Technologies and Mathematics

Information Communication Technologies (ICT) are powerful tools that can support student learning. Students can develop and demonstrate their understanding of concepts and content in Mathematics using a range of ICT tools. It is also important that students know how to use these ICT efficiently and responsibly, as well as learning how to protect themselves and secure their data.

Details of how ICT can support student learning in Mathematics is set out in the attached Information Communication Technologies and Mathematics pdf.

Scope and Sequence

The curriculum sets out what students are expected to learn and is designed as a continuum of learning. The curriculum is being presented in a scope and sequence chart to support teachers to easily see the progression and assist in planning teaching and learning programs to meet the diverse needs of students.

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These charts include the content descriptions and achievement standards. It is advised that these charts are read in conjunction with the introductory materials and the level/band descriptions in the curriculum.

The number of levels represented in each chart varies. Read the naming convention in the links below to assist in selecting the most appropriate chart.

These charts are designed to be printed in A3 format.

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Mathematics - F–10 Curriculum

Victorian Curriculum
Foundation–10

Level A

In Level A, students experience and respond to personally relevant and familiar situations and events that regularly and routinely involve activities and actions such as comparing, adding and removing, distributing, placing and moving.

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Level A Content Descriptions

Number and Algebra

Number and place value	Elaborations
Respond to objects being counted and distributed (VCMNA001)	• experiencing number names and number sequence in everyday experiences, for example birthdays, distributing equipment, collecting materials • experiencing number and its use in stories, songs and rhythm, chosen for their interest and relevance to the student's age • attending to number and counting in daily routines, for example cups of flour in the bowl, number of times an action or event is repeated or experienced • reacting to everyday situations where objects are counted, for example number of items on a plate or tray, steps to the door, eggs in the carton, bounces of a ball • being supported as they touch, feel or move over objects as they are being counted
Respond to situations where counting is involved (VCMNA002)	• experiencing to number name, numerals and quantities in everyday experiences • experiencing activities that use number names, numerals and quantities, for example stories, songs, cooking, music • attending to and being aware of number as objects are being counted • reacting to objects moving, disappearing and re-appearing within the context of everyday situations • responding to wanting more or less of an object
Respond to groups of personally relevant objects (VCMNA003)	• experiencing objects and groups of objects, for example through sensory activities • experiencing groups of objects of significantly different sizes • reacting to collections within the context of everyday familiar situations, for example collections of favourite food, jewellery, utensils, or collections based on colour or texture
Respond to situations where the comparison of two collections or objects is involved (VCMNA004)	• experiencing objects being collected and grouped together • using objects which are personally relevant to create collections • reacting to the comparison of everyday objects and collections of objects, for example using one-to-one correspondence to compare art equipment, cooking utensils, balls, sensory objects
Respond to the removal and addition of familiar items and objects in practical situations (VCMNA005)	• experiencing changes in quantity that occur during addition and subtraction within everyday experiences • attending to the adding and subtraction of familiar objects • reacting to changes in a collection when objects are added or subtracted
React to practical situations of sharing (VCMNA006)	• experiencing objects being distributed and assisting to distribute objects to others in everyday situations • reacting to objects being shared and distributed

Money and financial mathematics	Elaborations
React to everyday financial situations involving money (VCMNA007)	• being supported to experience the sensory characteristics of coins and notes • experiencing money being exchanged, counted and stored in various situations

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Patterns and algebra	Elaborations
Respond to the identification of objects (VCMNA008)	• experiencing patterns and the sorting of objects in everyday situations, for example repeated rhythm, pattern on textured material or objects, sorting of materials and utensils • becoming aware of ready-made visual, musical and multimodal patterns, for example, computer programs, stripes, dots, textures, sounds
Respond to repeated routines in everyday events (VCMNA009)	• experience steps to solve a problem • experience steps and sequences in everyday tasks • respond to repetition and repeated experiences within everyday life

Measurement and Geometry

Using units of measurement	Elaborations
Respond to objects based on length (VCMMG010)	• experiencing objects of various lengths • experiencing measurement attributes through the use of sensory input and objects • reacting to various measurement attributes of familiar objects, and being introduced to the names of those attributes, in the context of everyday experiences
Respond to personally relevant everyday events (VCMMG011)	• experiencing everyday events and their associated equipment and sensory elements • reacting to changes in their environment associated with everyday events • experiencing visuals being used to represent regular events • reacting to objects and tools associated with routine everyday events
Respond to personally relevant routine events (VCMMG012)	• experiencing events and their equipment, staff, and sensory elements • experiencing events and their characteristics being represented by images or augmentative and alternative communication (AAC) • reacting to routine events with the support of objects, sounds and textures • reacting and responding to daily events within the context of a range of practical situations, for example reacting to the bell for lunch time, responding to the timer, staff or an image for maths

Shape	Elaborations
Respond to familiar everyday shapes and objects (VCMMG013)	• experiencing a variety of objects and their characteristics, and being introduced to the names of shapes through interaction with objects • reacting to three-dimensional objects in everyday situations with the use of sensory input, for example touching, feeling, looking, shaking, rolling • responding to objects and shapes as they are identified in the classroom and environment in everyday situations

Location and transformation	Elaborations
Respond to movement of an object (VCMMG014)	• experiencing movement and position within various environments and everyday events, for example exploring the concept of ‘inside and outside’ during structured activities • experiencing the language used to name movement and changes in position during everyday experiences, such as up, down, over, under

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	• reacting to a variety of body positions and movements within the context of everyday activities • using everyday language of location and position to assist students to react to changes to objects and their body, for example, ‘arms up’, ‘I am moving your bag’, ‘Where did it go?’ • reaching for or looking for a desired object by moving head or body for a better view

Statistics and Probability	
Data representation and interpretation	Elaborations
Respond to objects relevant to a given context (VCMSP015)	• experiencing the collection, sorting and displaying of information and data, for example visual display of who is here today with photos of class members’ present • associating objects with familiar events, within the context of practical situations, for example recognising connection between own bag and lunchtime, equipment for art, sport, or cooking and the associated activity • experiencing the phenomena of chance by attending to activities and toys that demonstrate chance occurrences
Respond to objects being moved and organised to make a data display (VCMSP016)	• experiencing people's choices being organised into a simple display
Experience data display being interpreted (VCMSP017)	

Level A Achievement Standard

Number and Algebra

Students observe the use of number within their daily life. They begin to respond to numbers in everyday experiences. Students demonstrate awareness of counting by responding to number rhymes, songs, stories and finger games. They experience and respond to ‘one for you, one for me’, ‘gone’, ‘no more left’ and ‘give me more’. Students participate in making piles, groups or bundles of familiar everyday objects and respond to objects being put together and taken apart.

Measurement and Geometry

Students observe and explore objects within daily life. They react and respond to objects and experience measurement attributes in practical situations. Students explore objects of varying weights, lengths, capacities and materials. They show an awareness of time and daily routine by responding to a signal from the teacher, and items being brought out or removed. Students respond to a signal from a timer, used to indicate the end of an activity. Students explore and respond to objects of varying textures, colours, sizes and shapes. Students explore space by moving and changing position and location, and respond to changes in position.

Statistics and Probability

Students observe objects and events within their daily life. Students begin to display a similar and predictable reaction to regular events. They respond to major changes to regular games and activities associated with chance, surprise and predictability, such as moving a switch to activate a toy.

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Level B

In Level B, students participate in everyday activities involving making simple correspondences between objects, and explore qualitative, quantitative and comparative concepts in action and language such as none, one, more, same, different, again and gather.

They recognise regularity in some events and that not all events are certain.

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Level B Content Descriptions

Number and Algebra

Number and place value	Elaborations
Use number names ‘one’, ‘two’ and ‘three’ in sequence to count in everyday situations (VCMNA018)	• assisting to count objects initially to three, by participating in activities, stories, songs and rhymes that involve number. This could include nodding as objects are counted, tapping an object as it is counted, tracking objects being counted, repeating some number names as objects are being counted • responding to number of names in everyday situations, for example waiting for counting down or up before starting an action, sound, activity; stepping, clapping, making a sound, tapping a switch for each number • exploring and using some number names by ‘repeating’ number name either verbally or through a communication system or augmentative and alternative communication (AAC)
Correspond ‘one’ with a single object (VCMNA019)	• showing understanding of object permanency, by tracking and finding an object • exploring ‘oneness’ and indicate one, for example one clap, experiencing one object, giving out materials one at a time, requesting one, hearing one sound or beat • distributing objects and materials with support, using basic one-to-one correspondence
Explore the concept of ‘none’, ‘one’ and ‘more’ (VCMNA020)	• recognising a change in a collection’s size within the context of familiar, everyday objects, for example a group of cars, toys, lollies, fruit, vegetables, coins, books, materials or tools • demonstrating an understanding of one by selecting, giving or taking one object • responding to questions about group composition (for example ‘Which group has more?’ ‘Which group has none?’) verbally or by indicating through augmentative and alternative communication, actions or eye gaze
Make comparison between items using appropriate language such as ‘same’ or ‘different’ (VCMNA021)	• comparing two collections using one to one correspondence within the context of everyday situations • responding to questions about group comparison for example ‘Are they the same?’ ‘Or different?’ verbally or by indicating through augmentative and alternative communication, actions or eye gaze
Participate in everyday situations involving ‘adding’ and ‘taking away’ (VCMNA022)	• combining two groups of objects to make ‘more’ within the context of familiar everyday situations • participating in everyday situations that involve moving objects to make ‘less’ or ‘more’, for example dragging an object to or from a group, using a switch to initiate the action or augmentative and alternative communication to communicate the action required
Respond to everyday practical situations of sharing (VCMNA023)	• responding to the prompt to “share” by giving an object to another • assisting to hand out materials so everyone has the same materials

Money and financial mathematics	Elaborations
Respond to everyday financial situations involving money and match notes and coins (VCMNA024)	• matching coins or notes one to one which are 'the same'

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Patterns and algebra	Elaborations
Participate in the comparison of objects, using language such as 'same' and 'different' (VCMNA025)	• recognising patterns through the concept of 'same' and 'different' objects within the context of everyday situations • repeating rhythm patterns (with movement or sound) • making simple patterns by stacking and lining up objects • sorting everyday familiar objects during shared experiences, for example, when cooking, playing sport or playing with toys such as blocks
Follow a simple cause and effect process (VCMNA026)	• following routine everyday visual steps to solve everyday problem • explore cause and effects through play

Measurement and Geometry

Using units of measurement	Elaborations
Compare objects using direct comparison (VCMMG027)	• exploring objects in structured situations based on length • matching objects that are from a small field and identical in length • showing recognition or preference for an object based on its mass, capacity or length within a range of situations
Recognise and participate in familiar events that happen on a daily basis (VCMMG028)	• recognising the sequence of familiar events, for example 'first and then', with the support of picture-based schedules • understanding and using environmental cues such as equipment, staff and location to predict and identify activities • assisting in the construction of visual timetable for the class • responding to the signal from a timer to indicate the end of an activity
Participate in regular daily events (VCMMG029)	• developing an awareness of regular events and routines, with the support of daily and weekly visual schedules • recognising and following the routines at the start and end of the day

Shape	Elaborations
Identify when two shapes or objects are the same sort or not (VCMMG030)	• using 'posting' of items into different containers or holes to manipulate and sort shapes and objects • recognising three dimensional objects when identified in the environment using everyday situations • matching everyday objects that are 'the same' • exploring shapes by using 'play dough' and other malleable materials to make shapes • developing an awareness of the language used to describe shapes

Location and transformation	Elaborations
Respond to a simple statement about location or direction (VCMMG031)	• following directional communication used in daily routine, for example, 'put in' to finish an activity, 'sit down', 'stand up' during an activity or program • following everyday language of location and direction through modelled daily routines, for example: looking upwards for 'up'; using rhymes and chants that relate to movement; following a warm-up sequence for work or sport; unpacking their bag; washing their hands and setting up for lunch • exploring the concept of 'inside' and 'outside' during activities by putting objects

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	'in' and 'out' of a container and following routines and instructions that involve moving inside and outside locations

Statistics and Probability	
Data representation and interpretation	Elaborations
Participate in data collection (VCMSP032)	• following the construction of data display, for example picture representations of student choices, presented as class display of hair colour, favourite colour, song, movie, character • assisting to identify pictures that represent daily routines and events in a data display • choosing pictures or objects to put in a data display
Participate in the grouping of data (VCMSP033)	• answering a yes/no question and assisting to place their response in a data display • experiencing the representation of everyday situations and choices in simple date displays • match a response to a given category
Experiencing data being used for decision making in everyday situations (VCMSP034)	• experiencing data being totalled and read • experiencing data being used to identify the most popular choice and to make decisions on choices such as the type of transport they will use, animals they will visit first at the zoo, what book the class will read

Level B Achievement Standard

Number and Algebra

Students participate in everyday activities that involve numbers and counting, comparing groups of objects, and pattern activities. Students can rote count to three. Students identify 'one' and 'lots' of objects and show an understanding of 'more' in familiar situations. They manipulate objects and build a tall tower by using 'more' blocks and take blocks away from a tower to make the tower 'less' tall.

Measurement and Geometry

Students participate in everyday activities that explore measurement and use measurement attributes in practical situations. Students demonstrate beginning understanding of basic measurement concepts such as 'long or short', 'heavy or light'. They explore routine events and show an awareness of time and daily routines by responding to a routine signal from the teacher. They demonstrate an awareness of object permanence by searching for objects that have been hidden and participate in class activities that explore three-dimensional objects. They can match identical familiar three-dimensional shapes that are 'the same'. Students respond to specific instructions relating to manipulating the movement and location of self and objects.

Statistics and Probability

Students participate in class activities that explore object, events and displaying information. They develop an awareness of chance by playing with materials or objects that involve cause and effect (actions that will happen) and playing games where the outcome is unpredictable. Students respond to a simple pictorial representation of their activities related to a short time-frame.

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Level C

In Level C, students demonstrate awareness in practical situations, connecting objects, numbers names and numerals from one to three, using ‘one more than’ and ‘one less than’.

They match like objects, describe differences between objects for given attribute, show familiarity with stages of time and events within a day using visual schedules, visually locate named objects in a familiar environment and identify some data relevant to a situation.

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Level C Content Descriptions

Number and Algebra

Number and place value	Elaborations
Use number names in sequence to count in everyday situations, initially from one to five (VCMNA035)	• developing fluency in forward counting to five using words, sign or augmentative and alternative communication (AAC) in meaningful contexts, including everyday experiences and stories • knowing that the forward sequence of counting words occurs in the same order, for example always communicating numbers in the same sequence, indicating when another person uses number names in the wrong sequence • developing one-to-one correspondence by recognising that each object is counted, by indicating an object during counting situations, for example, indicating by pointing, activating a switch or eye gaze
Know and match number name, numerals and quantities to three (VCMNA036)	• developing one-to-one matching of number word or its representation through sign or alternative and augmentative communication (AAC) to objects initially up to three • recognising that numerals look different from non-numeral shapes • using structured situations to count and match groups of objects to a numeral, initially up to 3
Identify groups as being 'one', 'more' or 'less' (VCMNA037)	• indicating the larger of two groups when presented with groups of significantly different number sets, for example group of paints, lollies, tools, dots, blocks, counters
Compare and order two collections according to their quantity (VCMNA038)	• comparing and ordering items using appropriate language of 'more', 'less', 'same' • using everyday situations of one-to-one correspondence to compare two collections of significantly different amounts • using practical situations to indicate 'more', 'less' or 'same'
Demonstrate in practical situations, 'adding one more to' and 'taking one away from' in everyday situations (VCMNA039)	• using shared experiences with concrete materials to add one more to or take away one from a group of objects, and count to find a total ○ experiencing addition and subtraction in games • communicating using language such as 'more' or 'less' to describe a change in a group using classroom resources and in practical situations
Sharing materials in practical situations (VCMNA040)	• using one to one correspondence to distribute materials evenly • exploring the distribution of a group of concrete materials into different numbers of groups

Money and financial mathematics	Elaborations
Use direct comparison to sort coins or notes into groups (VCMNA041)	• identify and group notes or coins that are 'the same' • group coins based on a set criteria, for example size, colour, shape, coins, notes

Patterns and algebra	Elaborations
Pair identical objects from a small collection, and recognise simple repeated patterns (VCMNA042)	• copying a simple repeated pattern using one-to-one correspondence with objects • copying a pattern associated with a familiar activity, for example repeating movement pattern

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	• using a single given attribute (for example, size, colour, texture, shape) to group objects • matching, sorting and organising objects in practical situations
Identify repeated routines and sequences in everyday events (VCMNA043)	• follow and assist to represent a sequence of steps • follow a simple decision (algorithm) to solve a everyday problem • participate in a cause and effects activities, games and events

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Measurement and Geometry

Using units of measurement	Elaborations
Compare two objects based on measurement attributes of length (VCMMG044)	• comparing and matching objects based on their having the 'same' or 'different' length, for example pencils, shoes or food items that are the same length • recognising length in shared experiences by comparing objects directly and indicating differences between objects • using measurement language such as long or short to communicate significant differences between objects
Identify familiar events that occur at different stages of a day (morning, afternoon, evening, night) (VCMMG045)	• sorting images or symbols according to whether they occur during the day or the night • using visual schedules to complete activities within an event and events within the day, and manage time throughout the day • interpreting language, images and communication associated with day such as morning, afternoon, and night • indicating an event as long or short • recognising and responding to daily routines by selecting an image from a schedule to indicate an activity has finished or is about to begin
Identify the familiar events within the day using a visual schedule (VCMMG046)	• identifying daily events through materials, visuals and objects provided • identifying daily events with the assistance of visual or tactile schedules and timetables • identifying events that may or may not happen today

Shape	Elaborations
Match familiar two dimensional shapes and three dimensional objects (VCMMG047)	• matching simple three dimensional shapes with similar attributes, for example, different-sized spheres ○ matching simple two dimensional shapes with similar attributes, for example, different-sized spheres • recognising the relationship between familiar shapes using insert puzzles

Location and transformation	Elaborations
Locate familiar three-dimensional objects in the classroom when they are named (VCMMG048)	• interpreting everyday language of location by following simple directions and communication, for example, move or track an object, feel or follow a path • locating familiar environments and objects, for example, classroom, hall, desk

Statistics and Probability

Data representation and interpretation	Elaborations
Identify data relevant to a given context (VCMSP049)	• assisting to collect information to answer a question, for example, what colours are the counters in the box • following simple instructions to sort objects for a data display
Follow simple instructions to sort objects into a simple data display (VCMSP050)	• making and collecting people's choices and assisting to collate these in a data display based on set options • match responses to establish groups to assist in collating data

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Identify the choices/responses of a data display (VCMSP051)	• indicate their eye colour from a display, indicate who is here or not here

Level C Achievement Standard

Number and Algebra

Students connect number names and numerals with sets of up to five elements. They match individual objects with counting sequences up to and back from five. Students use concrete materials to solve problems that involve comparing, combining and separating sets. Students make ‘groups’, ‘lots’ and groups of ‘one’ and can indicate which collection has ‘more’ than the other. They can distribute objects to each person in a group until there are no objects left. Students order the first three elements of a set. Students can match one attribute of familiar objects.

Measurement and Geometry

Students explore measurement attributes in practical situations and use words to describe the characteristics of familiar objects. Students solve simple mathematical problems associated with longer and shorter lengths. They explore events and identify day and night events. They can identify events that may or may not happen today. Students respond to a simple pictorial representation of activities related to their whole day. They match objects that are the same and sort familiar objects, and an understanding of the concept of ‘inside and outside’ by following instructions. They demonstrate an understanding of location and spatial awareness by following simple instructions related to simple spatial concepts.

Statistics and Probability

Students participate in and contribute to the development of picture schedules, timetables and pictorial lists associated with familiar activities, such as listing the ingredients needed for a cooking session. They demonstrate an understanding of the concept of chance by participating in games of chance, and identifying events that may or may not happen today.

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Level D

In Level D, students actively use concrete models to represent number in various situations, and use number names when comparing, counting, ordering, adding to or taking away from sets of one to five objects.

They form the initial terms of basic patterns and identify and sequence regular daily events, make simple qualitative measurement comparisons, match objects with their pictorial representations, and name them.

Students follow instructions to place and find objects in various locations and they recognise that similar events or activities may have different outcomes at different times.

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Level D Content Descriptions

Number and Algebra

Number and place value	Elaborations
Use a number names in sequence to count in everyday situations, initially from one to ten (VCMNA052)	• developing fluency with forward counting by communicating number names while counting, initially to 10 and in meaningful contexts such as cooking, collecting equipment, stories and games • understanding one-to-one correspondence by knowing that each object is counted only once, by tracking an object while counting in shared and structured counting experiences, for example moving objects once counted, counting objects left to right
Recognise number name, numerals and quantities, initially up to five and beyond (VCMNA053)	• responding to key vocabulary and questions about 'how many' • using one-to-one matching of number words, sign or augmentative and alternative communication (AAC) representation for objects to five • matching numerals to the correct number of items initially to five using number games, software, cards and everyday situations
Subitise regular arrangements of objects and arrays up to five (VCMNA054)	• understanding that some numbers are represented by a set pattern or array
Compare, order and make comparisons between two collections, according to their quantity, using numbers initially to five (VCMNA055)	• comparing and ordering collections using the appropriate language and number name • comparing and communicating about characteristics of groups, for example sequencing objects in a group from smallest to largest; indicating the larger/smaller group through gesture, verbally or augmentative and alternative communication (AAC)
Model practical situations involving 'adding to' or 'taking away' with collections of up to five objects (VCMNA056)	• using shared experiences with concrete materials to combine two groups of objects, and count to find a total • counting on or back from a group using concrete materials, for example adding/ subtracting balls in a basket, pens in a container, tools in a tool box • using a calculator or digital device to undertake simple addition and subtraction • exploring the concept of adding one and taking away one
Sharing material in practical situations so everyone has the same amount (VCMNA057)	• using one to one correspondence to share materials equally • exploring stories and everyday situations which require sharing and the distributing of materials

Money and financial mathematics	Elaborations
Using money in everyday financial situations and matching coins to two dimensional images (VCMNA058)	• matching coins to 2D images • using money to pay for goods and services in the community

Patterns and algebra	Elaborations
Sort like objects based on a given classification, and identify and continue a simple repeated pattern	• continuing a simple repeated two-part pattern with given objects, movements or sounds, for example, red, blue, red (set the table with drinking cups), paper, glue (set up equipment), knife and fork (set a table using two implements)

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with its next element (VCMNA059)	• making patterns and collections based on objects being the same, different and alike • examining and continuing a simple repeated pattern with given objects, movements or sounds, for example, continue a two-beaded pattern (red, blue, red, blue) • sorting and grouping objects and communicating attributes such as texture, colour, size, category; for example, sort papers into two colours, sort cutlery into two groups • recognising patterns in familiar environments
Follow a sequence of steps (VCMNA060)	• using sequence steps to complete an activity • representing a sequence of steps using visuals • identify options and following steps through everyday situations and events • participate in a variety of chance games and experiences

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Measurement and Geometry

Using units of measurement	Elaborations
Respond to contexts involving 'heavier/lighter' than and 'holds more/less' than (VCMMG061)	• using measurement language such as longer and shorter, or heavier and lighter, to communicate differences between objects • using direct comparison to compare objects based on their length, mass or volume
Identify and sequence regular events that occur during the school day and comment on their duration (short/long) (VCMMG062)	• creating and following a visual schedule based on the school day's events • ordering events based on duration, for example, 'I can do more at lunch time than recess' • comparing and labelling events as long, short, quick • understanding the purpose of a clock and some of its features • using timers and a clock to explore and compare time and duration of events
Identify the days of the week in sequence (VCMMG063)	• using picture schedules to identify key events of a day and a week • using everyday language 'sometimes', 'always' and never to indicate the likelihood of an event on a day or week • using 'morning routine' to identify the days and what will occur during the day • communicating the days of the week

Shape	Elaborations
Use direct comparison to sort three dimensional objects and two dimensional shapes (VCMMG064)	• sorting shapes that are the 'same' or 'different' • identifying, matching and sorting simple familiar two and three dimensional objects according to a single attribute • identifying familiar shapes hidden in a picture • investigating the inside and outside shape of an object • understanding the relationship between shape and template, for example, insert puzzle, inserting a shape in a matching shaped hole

Location and transformation	Elaborations
Follow simple directional words to locate or move an object 'on', 'in' or 'under' (VCMMG065)	• following multiple positional directions during everyday situations, for example, chants, songs or rhymes with repeated actions, completing an art, cooking or drawing activity, locating a tool or completing a job • developing consistency and fluency in communicating and interpreting directional and positional words • using everyday location language to explain where an object is

Statistics and Probability

Data representation and interpretation	Elaborations
Answer simple yes/no questions about data that has been gathered in a given context (VCMSP066)	• following and constructing a data display such as routine schedule or daily timetable • collecting and displaying data in a personally meaningful way
Collect and display data in response to a question using materials	• representing responses to everyday situations with everyday objects into data displays • collate responses to a question into groups

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(VCMSP067)	• match responses which are the same to assist in data collation
Identify what the data display is representing and answer questions using yes/no responses (VCMSP068)	• identifying which response is the largest • identify which response has the least

Level D Achievement Standard

Number and Algebra

Students connect number names and numerals with sets of up to 10 elements. They match individual objects with counting sequences up to and back from 10. They recognise and point to numerals in and around the classroom, for example, numbers on a clock face. Students use concrete materials to solve problems that involve comparing, combining and separating sets. They can indicate when groups of less than 10 objects are the same or different in number and that two collections have the ‘same’ quantity by matching items one to one. They can find the first and last object in a sequence and place objects into sets to make ‘more’ and take objects from a group to make ‘less’. Students order the first five elements of a set. They sort objects and shapes based on a given attribute and create simple repeating patterns of two elements or more by copying a pattern.

Measurement and Geometry

Students explore measurement attributes in practical situations and identify and describe the basic characteristics of a range of objects. They can identify regular events within the school week. They can follow a class pictorial schedule and mark off each passing day on a calendar. Students demonstrate an understanding of two- and three-dimensional shapes by matching basic geometric objects to pictures of that object, identifying basic three-dimensional shapes in the classroom and sorting shapes into like groups. Students show an understanding of ‘location’ and spatial concepts by responding to instructions to position items.

Statistics and Probability

Students explore events and follow a simple picture schedule, and use these to answer simple ‘yes’ or ‘no’ questions. They play a variety of chance games such as bingo or snakes and ladders and demonstrate an understanding that they will not always win.

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Foundation Level

In Foundation level, students play with objects and draw pictures to develop links between their immediate environment, everyday language and mathematical activity.

Students classify and sort objects into sets and form simple correspondences between them. They decide when two sets are of equal size, or one is smaller or bigger than another. They develop an understanding of the concepts of number and numeral, count, order, add and share using small sets of objects. They create and continue simple patterns.

Students compare common objects with respect to length, mass and capacity, and order events and compare their duration. They make rough estimates and simple measurements with respect to informal units. Students name, sort and describe familiar everyday shapes and objects, and describe position and movement in their immediate environment.

Students investigate situations requiring data collection and presentation in simple displays, and recognise unpredictability and uncertainty in some events.

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Foundation Level Content Descriptions

Number and Algebra

Number and place value	Elaborations
Establish understanding of the language and processes of counting by naming numbers in sequences, initially to and from 20, moving from any starting point (VCMNA069)	• reading stories from other cultures featuring counting in sequence to assist students to recognise ways of counting in local languages and across cultures • identifying the number words in sequence, backwards and forwards, and reasoning with the number sequences, establishing the language on which subsequent counting experiences can be built • developing fluency with forwards and backwards counting in meaningful contexts, including stories and rhymes • understanding that numbers are said in a particular order and there are patterns in the way we say them
Connect number names, numerals and quantities, including zero, initially up to 10 and then beyond (VCMNA070)	• understanding that each object must be counted only once, that the arrangement of objects does not affect how many there are, and that the last number counted answers the 'how many' question • using scenarios to help students recognise that other cultures count in a variety of ways, such as by placing one pebble in a bag to represent one object. For example to count the number of cattle
Subitise small collections of objects (VCMNA071)	• using subitising as the basis for ordering and comparing collections of numbers
Compare, order and make correspondences between collections, initially to 20, and explain reasoning (VCMNA072)	• comparing and ordering items of like and unlike characteristics using the words 'more', 'less', 'same as' and 'not the same as' and giving reasons for these answers • understanding and using terms such as 'first' and 'second' to indicate ordinal position in a sequence. • using objects which are personally and culturally relevant to students
Represent practical situations to model addition and subtraction (VCMNA073)	• using a range of practical strategies for adding and subtracting small groups of numbers, such as visual displays or concrete materials • using Aboriginal and Torres Strait Islander methods of adding and subtracting, including spatial patterns and reasoning
Represent practical situations to model sharing (VCMNA074)	• using a range of practical strategies for sharing small groups of numbers, such as visual displays or concrete materials • using verbal action stories to model situations that involve sharing

Money and financial mathematics	Elaborations
Represent simple, everyday financial situations involving money (VCMNA075)	• using toy money to pay for goods in play situations

Patterns and algebra	Elaborations
Sort and classify familiar objects and explain the basis for these classifications, and copy, continue and create patterns with objects and	• observing natural patterns in the world around us • creating and describing patterns using materials, sounds, movements or drawings • extending patterns using materials and drawings to the right and to the left

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drawings (VCMNA076)	• identifying which part of the pattern is being repeated (happening over and over again)
Follow a short sequence of instructions (VCMNA077)	• carrying out a specified sequence of actions to move an object from one location to another • playing a simple rule-based game moving a specified number of places according to the result on a die in a chance-based game

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Measurement and Geometry

Using units of measurement	Elaborations
Use direct and indirect comparisons to decide which is longer, heavier or holds more, and explain reasoning in everyday language (VCMMG078)	• comparing objects directly, by placing one object against another to determine which is longer or by pouring from one container into the other to see which one holds more • using suitable language associated with measurement attributes, such as ‘tall’ and ‘taller’, ‘heavy’ and ‘heavier’, ‘holds more’ and ‘holds less’
Compare and order the duration of events using the everyday language of time (VCMMG079)	• knowing and identifying the days of the week and linking specific days to familiar events • sequencing familiar events in time order
Connect days of the week to familiar events and actions (VCMMG080)	• choosing events and actions that make connections with students’ everyday family routines

Shape	Elaborations
Sort, describe and name familiar two-dimensional shapes and three-dimensional objects in the environment (VCMMG081)	• sorting and describing squares, circles, triangles, rectangles, spheres and cubes

Location and transformation	Elaborations
Describe position and movement (VCMMG082)	• interpreting the everyday language of location and direction, such as ‘between’, ‘near’, ‘next to’, ‘forwards’, ‘towards’ • following and giving simple directions to guide a friend around an obstacle path and vice versa

Statistics and Probability

Data representation and interpretation	Elaborations
Answer yes/no questions to collect information (VCMSP083)	• posing questions about themselves and familiar objects and events, for example, 'Do you have any pets?' • answering questions that have exactly two mutually exclusive possible responses
Organise answers to yes/no questions into simple data displays using objects and drawings (VCMSP084)	• representing responses to questions using simple displays, including grouping students according to their answers
Interpret simple data displays about yes/no questions (VCMSP085)	• using data displays to answer simple questions such as ‘how many students answered “yes” to having pets?’

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Foundation Level Achievement Standard

Number and Algebra

Students connect number names and numerals with sets of up to 20 elements, estimate the size of these sets, and use counting strategies to solve problems that involve comparing, combining and separating these sets. They match individual objects with counting sequences up to and back from 20. Students order the first 10 elements of a set. They represent, continue and create simple patterns.

Measurement and Geometry

Students identify measurement attributes in practical situations and compare lengths, masses and capacities of familiar objects. They order events, explain their duration, and match days of the week to familiar events. Students identify simple shapes in their environment and sort shapes by their common and distinctive features. They use simple statements and gestures to describe location.

Statistics and Probability

Students sort familiar categorical data into sets and use these to answer yes/no questions and make simple true/false statements about the data.

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Level 1

In Level 1, students use mathematical symbols and language as well as materials and drawings in their mathematical explorations of daily life.

Students recognise, represent and order numbers to at least 100 using materials, diagrams, words, numerals and a number line, and apply this with respect to the value of Australian coins. They group and skip count by twos, fives and tens, and count to 100 by partitioning and using place value. Students solve simple addition problems, and share into two equal groups or parts to model one-half.

Students use uniform informal units to measure and compare length and capacity. They tell time to the half-hour and use time and calendar terms such as hours, days, weeks and months to describe duration. Students use terms such as corner, edge and face to classify familiar shapes and objects, and are able to give and follow directions to familiar locations.

Students use one-to-one correspondences to display categorical data obtained from a simple investigation. They identify chance events in familiar contexts and use everyday language such as ‘will happen’, won’t happen’ or ‘might happen’ in relation to these.

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Level 1 Content Descriptions

Number and Algebra

Number and place value	Elaborations
Develop confidence with number sequences to and from 100 by ones from any starting point. Skip count by twos, fives and tens starting from zero (VCMNA086)	• using the popular Korean counting game sam-yuk-gu for skip counting • developing fluency with forwards and backwards counting in meaningful contexts such as circle games
Recognise, model, read, write and order numbers to at least 100. Locate these numbers on a number line (VCMNA087)	• modelling numbers with a range of material and images • identifying numbers that are represented on a number line and placing numbers on a prepared number line
Count collections to 100 by partitioning numbers using place value (VCMNA088)	• understanding partitioning of numbers and the importance of grouping in tens • understanding two-digit numbers as being comprised of tens and ones/units
Represent and solve simple addition and subtraction problems using a range of strategies including counting on, partitioning and rearranging parts (VCMNA089)	• developing a range of mental strategies for addition and subtraction problems
Represent practical situations that model sharing (VCMNA090)	• sharing a set of objects, such as a packet of sweets, equally between a small group of people using one-to-one correspondence • recognising whether there is a remainder or not after sharing equally

Fractions and decimals	Elaborations
Recognise and describe one-half as one of two equal parts of a whole (VCMNA091)	• sharing a collection of readily available materials into two equal portions • splitting an object into two equal pieces and describing how the pieces are equal

Money and financial mathematics	Elaborations
Recognise, describe and order Australian coins according to their value (VCMNA092)	• showing that coins are different in other countries by comparing Asian coins to Australian coins • understanding that the value of Australian coins is not related to size • describing the features of coins that make it possible to identify them

Patterns and algebra	Elaborations
Investigate and describe number patterns formed by skip counting and patterns with objects (VCMNA093)	• using place-value patterns beyond the teens to generalise the number sequence and predict the next number • investigating patterns in the number system, such as the occurrence of a particular digit in the numbers to 100

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Recognise the importance of repetition of a process in solving problems (VCMNA094)	• using one-to-one correspondence to determine which of two sets is larger, or if they are of equal size • dividing a set of blocks in a simple ratio such as ‘2 for me’, ‘1 for you’

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Measurement and Geometry

Using units of measurement	Elaborations
Measure and compare the lengths, masses and capacities of pairs of objects using uniform informal units (VCMMG095)	• understanding that in order to compare objects, the unit of measurement must be the same size • lifting to compare the mass of objects using words, for example, heavier, lighter, same • measuring the capacity of containers using uniform material, for example cups or buckets
Tell time to the half-hour (VCMMG096)	• reading time on analogue and digital clocks and observing the characteristics of half-hour times
Describe duration using months, weeks, days and hours (VCMMG097)	• describing the duration of familiar situations such as ‘how long is it until we next come to school?’

Shape	Elaborations
Recognise and classify familiar two-dimensional shapes and three-dimensional objects using obvious features (VCMMG098)	• focusing on geometric features and describing shapes and objects using everyday words such as 'corners', 'edges' and 'faces'

Location and transformation	Elaborations
Give and follow directions to familiar locations (VCMMG099)	• understanding that people need to give and follow directions to and from a place, and that this involves turns, direction and distance • understanding the meaning and importance of words such as ‘clockwise’, ‘anticlockwise’, ‘forward’ and ‘under’ when giving and following directions • interpreting and following directions around familiar locations

Statistics and Probability

Chance	Elaborations
Identify outcomes of familiar events involving chance and describe them using everyday language such as ‘will happen’, ‘won’t happen’ or ‘might happen’ (VCMSP100)	• justifying that some events are certain or impossible

Data representation and interpretation	Elaborations
Choose simple questions and gather responses (VCMSP101)	• determining which questions will gather appropriate responses for a simple investigation

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Represent data with objects and drawings where one object or drawing represents one data value. Describe the displays (VCMSP102)	• understanding one-to-one correspondence • describing displays by identifying categories with the greatest or least number of objects

Level 1 Achievement Standard

Number and Algebra

Students count to and from 100 and locate these numbers on a number line. They partition numbers using place value and carry out simple additions and subtractions, using counting strategies. Students recognise Australian coins according to their value. They identify representations of one half. Students describe number sequences resulting from skip counting by 2s, 5s and 10s. They continue simple patterns involving numbers and objects with and without the use of digital technology.

Measurement and Geometry

Students use informal units of measurement to order objects based on length, mass and capacity. They tell time to the half-hour and explain time durations. Students describe two-dimensional shapes and three-dimensional objects. They use the language of distance and direction to move from place to place.

Statistics and Probability

Students describe data displays. They ask questions to collect data and draw simple data displays. Students classify outcomes of simple familiar events.

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Level 2

In Level 2, students use grouping partitioning and re-arrangement to apply place value and extend the range of numbers they use and apply to thousands.

Students recognise, model and order numbers to at least 1000 and use a variety of strategies to count efficiently, including skip counting forwards and backwards by twos threes, fives and tens, with and without the use of technology. They explore the relationship between addition and subtraction, and use a variety of strategies to solve problems, including missing number problems. Students use groups and arrays to represent multiplication and division and solve simple problems, including finding halves, quarters and eighths of sets and shapes. They count and order by value, small collections of Australian coins and notes.

Students compare and order sets of shapes and objects based on length, area, volume and capacity using uniform informal units. They compare masses using balance scales, tell the time to the quarter hour, and use months and seasons to describe sequences of events over a longer time frame. Students describe sets of shapes and objects defined in terms of properties, and draw examples of these with and without the use of technology. They use simple maps and identify relative locations, and investigate the effect of simple transformations of slides, flips, half and quarter turns, both by hand and using technology.

Students use questions of interest to gather and display data for a single categorical variable and interpret it. They identify chance in a range of activities and describe related outcomes as ‘likely’ or ‘unlikely’.

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Level 2 Content Descriptions

Number and Algebra

Number and place value	Elaborations
Investigate number sequences, initially those increasing and decreasing by twos, threes, fives and ten from any starting point, then moving to other sequences (VCMNA103)	• developing fluency and confidence with numbers and calculations by saying number sequences • recognising that the natural numbers with zero form an ordered infinite set {0, 1, 2, 3 … } with a first element but no last element • recognising patterns in number sequences, such as adding 10 always results in the same final digit
Recognise, model, represent and order numbers to at least 1000 (VCMNA104)	• recognising there are different ways of representing numbers and identifying patterns going beyond 100 • developing fluency with writing numbers in meaningful contexts
Group, partition and rearrange collections up to 1000 in hundreds, tens and ones to facilitate more efficient counting (VCMNA105)	• using an abacus to model and represent numbers • understanding three-digit numbers as comprised of hundreds, tens and ones/units • demonstrating and using models such as linking blocks, sticks in bundles, place-value blocks and Aboriginal bead strings and explaining reasoning
Explore the connection between addition and subtraction (VCMNA106)	• becoming fluent with partitioning numbers to understand the connection between addition and subtraction • using counting on to identify the missing element in an additive problem
Solve simple addition and subtraction problems using a range of efficient mental and written strategies (VCMNA107)	• becoming fluent with a range of mental strategies for addition and subtraction problems, such as commutativity for addition, building to 10, doubles, 10 facts and adding 10 • modelling and representing simple additive situations using materials such as 10 frames, 20 frames and empty number lines
Recognise and represent multiplication as repeated addition, groups and arrays (VCMNA108)	• representing array problems with available materials and explaining reasoning • visualising a group of objects as a unit and using this to calculate the number of objects in several identical groups
Recognise and represent division as grouping into equal sets and solve simple problems using these representations (VCMNA109)	• dividing the class or a collection of objects into equal-sized groups • identifying the difference between dividing a set of objects into three equal groups (partition) and dividing the same set of objects into groups of three (quotition)

Fractions and decimals	Elaborations
Recognise and interpret common uses of halves, quarters and eighths of shapes and collections (VCMNA110)	• recognising that sets of objects can be partitioned in different ways to demonstrate fractions • relating the number of parts to the size of a fraction

Money and financial mathematics	Elaborations

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Count and order small collections of Australian coins and notes according to their value (VCMNA111)	• identifying equivalent values in collections of coins or notes, such as two five-cent coins having the same value as one 10-cent coin • counting collections of coins or notes to make up a particular value, such as that shown on a price tag

Patterns and algebra	Elaborations
Describe patterns with numbers and identify missing elements (VCMNA112)	• describing a pattern created by skip counting and representing the pattern on a number line • investigating features of number patterns resulting from adding twos, fives or 10s
Solve problems by using number sentences for addition or subtraction (VCMNA113)	• representing a word problem as a number sentence • writing a word problem to represent a number sentence
Apply repetition in arithmetic operations, including multiplication as repeated addition and division as repeated subtraction (VCMNA114)	• using technology to construct a sequence of numbers based on constant addition or subtraction from a given starting value • sharing a set of objects equally between a small number of groups

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Measurement and Geometry

Using units of measurement	Elaborations
Compare and order several shapes and objects based on length, area, volume and capacity using appropriate uniform informal units (VCMMG115)	• comparing lengths using finger length, hand span or a piece of string • comparing areas using the palm of the hand or a stone • comparing capacities using a range of containers
Compare masses of objects using balance scales (VCMMG116)	• using balance scales to determine whether the mass of different objects is more, less or about the same
Tell time to the quarter-hour, using the language of 'past' and 'to' (VCMMG117)	• describing the characteristics of quarter-past times on an analogue clock, and identifying that the small hand is pointing just past the number and the big hand is pointing to the three
Name and order months and seasons (VCMMG118)	• investigating the seasons used by Aboriginal people, comparing them to those used in Western society and recognising the connection to weather patterns.
Use a calendar to identify the date and determine the number of days in each month (VCMMG119)	• using calendars to locate specific information, such as finding a given date on a calendar and saying what day it is, and identifying personally or culturally specific days

Shape	Elaborations
Describe and draw two-dimensional shapes, with and without digital technologies (VCMMG120)	• identifying key features of squares, rectangles, triangles, kites, rhombuses and circles, such as straight lines or curved lines, and counting the edges and corners
Describe the features of three-dimensional objects (VCMMG121)	• identifying geometric features such as the number of faces, corners or edges

Location and transformation	Elaborations
Interpret simple maps of familiar locations and identify the relative positions of key features (VCMMG122)	• understanding that we use representations of objects and their positions, such as on maps, to allow us to receive and give directions and to describe place • constructing arrangements of objects from a set of directions
Investigate the effect of one-step slides and flips with and without digital technologies (VCMMG123)	• understanding that objects can be moved but changing position does not alter an object's size or features
Identify and describe half and quarter turns (VCMMG124)	• predicting and reproducing a pattern based around half and quarter turns of a shape and sketching the next element in the pattern

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Statistics and Probability

Chance	Elaborations
Identify practical activities and everyday events that involve chance. Describe outcomes as 'likely' or 'unlikely' and identify some events as 'certain' or 'impossible' (VCMSP125)	• classifying a list of everyday events according to how likely they are to happen, using the language of chance, and explaining reasoning

Data representation and interpretation	Elaborations
Identify a question of interest based on one categorical variable. Gather data relevant to the question (VCMSP126)	• determining the variety of birdlife in the playground and using a prepared table to record observations
Collect, check and classify data (VCMSP127)	• recognising the usefulness of tally marks • identifying categories of data and using them to sort data
Create displays of data using lists, table and picture graphs and interpret them (VCMSP128)	• creating picture graphs to represent data using one-to-one correspondence • comparing the usefulness of different data displays

Level 2 Achievement Standard

Number and Algebra

Students count to and from, and order numbers up to 1000. They perform simple addition and subtraction calculations, using a range of strategies. They find the total value of simple collections of Australian notes and coins. Students represent multiplication and division by grouping into sets and divide collections and shapes into halves, quarters and eighths. They recognise increasing and decreasing number sequences involving 2s, 3s, 5s and 10s, identify the missing element in a number sequence, and use digital technology to produce sequences by constant addition.

Measurement and Geometry

Students order shapes and objects, using informal units for a range of measures. They tell time to the quarter hour and use a calendar to identify the date, days, weeks and months included in seasons and other events. Students draw two-dimensional shapes, specify their features and explain the effects of one-step transformations. They recognise the features of three-dimensional objects. They interpret simple maps of familiar locations.

Statistics and Probability

Students collect data from relevant questions to create lists, tables and picture graphs with and without the use of digital technology. They interpret data in context. Students use everyday language to describe outcomes of familiar events.

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Level 3

In Level 3, students increasingly use mathematical terms and symbols to describe computations, measurements and characteristics of objects.

Students recognise, model and order numbers to at least 10 000 and place four digit numbers on a number line with regard for scale. They partition and re-arrange to facilitate calculations involving addition and subtraction. Students have facility with single digit addition and related subtraction facts, and recall multiplication and related division facts for twos, threes, fives and tens. They formulate and solve simple multiplication and division problems, estimate answers and use technology to check calculations. Students group money to a specified value in several ways, and calculate change required in simple transactions. They model and represent multiples of unit fractions up to a whole, using arrays on a number line. They write simple rules for number patterns and generate those patterns.

Students use metric units of length, mass and capacity to measure, order and compare objects. They associate angle with measure of turn and compare angles in everyday situations. They tell the time in minutes and convert between units of time. They use simple grids in maps and identify symmetry.

Students carry out investigations, collect and organise data into categories and use different methods with and without technology to display the data. They conduct experiments involving chance, describe possible outcomes and recognise variability in results.

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Level 3 Content Descriptions

Number and Algebra

Number and place value	Elaborations
Investigate the conditions required for a number to be odd or even and identify odd and even numbers (VCMNA129)	• identifying even numbers using skip counting by twos or by grouping even collections of objects in twos • explaining why all numbers that end in the digits 0, 2, 4, 6 and 8 are even and that numbers ending in 1, 3, 5, 7 and 9 are odd
Recognise, model, represent and order numbers to at least 10 000 (VCMNA130)	• placing four-digit numbers on a number line using an appropriate scale • reproducing numbers in words using their numerical representations and vice versa
Apply place value to partition, rearrange and regroup numbers to at least 10 000 to assist calculations and solve problems (VCMNA131)	• recognising that 10 000 equals 10 thousands, 100 hundreds, 1000 tens and 10 000 ones • justifying choices about partitioning and regrouping numbers in terms of their usefulness for particular calculations
Recognise and explain the connection between addition and subtraction (VCMNA132)	• demonstrating the connection between addition and subtraction using partitioning or by writing equivalent number sentences • solving simple word problems involving addition or subtraction
Recall addition facts for single-digit numbers and related subtraction facts to develop increasingly efficient mental strategies for computation (VCMNA133)	• recognising that certain single-digit number combinations always result in the same answer for addition and subtraction, and using this knowledge for addition and subtraction of larger numbers • extending strategies for addition and subtraction such as 14 + 8 + 6 = 14 + 6 + 8 = 28 and 54 – 28 = 2 + 20 + 4 • combining knowledge of addition and subtraction facts and partitioning to aid computation. For example, 57 + 19 = 57 + 20 – 1
Recall multiplication facts of two, three, five and ten and related division facts (VCMNA134)	• establishing multiplication facts using number sequences • using strategies to recall the multiplication and related division facts for the twos, threes, fives and tens
Represent and solve problems involving multiplication using efficient mental and written strategies and appropriate digital technologies (VCMNA135)	• writing simple word problems in numerical form and vice versa • using technology to check the solution and reasonableness of the answer

Fractions and decimals	Elaborations
Model and represent unit fractions including 1/2, 1/4, 1/3, 1/5 and their multiples to a complete whole (VCMNA136)	• partitioning areas, lengths and collections to create halves, thirds, quarters and fifths, such as folding the same sized sheets of paper to illustrate different unit fractions and comparing the number of parts with their sizes • locating unit fractions on a number line • recognising that in English the term ‘one third’ is used (order: numerator, denominator) but in other languages, such as Japanese for example, this concept may be expressed as ‘three parts, one of them’ (order: denominator, numerator)

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Money and financial mathematics	Elaborations
Represent money values in multiple ways and count the change required for simple transactions to the nearest five cents (VCMNA137)	• recognising the relationship between dollars and cents, and that not all countries use these denominations and divisions. For example, the Japanese Yen

Patterns and algebra	Elaborations
Describe, continue, and create number patterns resulting from performing addition or subtraction (VCMNA138)	• identifying and writing the rules for number patterns • describing a rule for a number pattern, then creating the pattern
Use a function machine and the inverse machine as a model to apply mathematical rules to numbers or shapes (VCMNA139)	• finding and describing simple rules in words to solve problems • using simple function machines to represent and apply a process or the inverse process, such as increase or decrease the value of a number by a specified amount

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Measurement and Geometry

Using units of measurement	Elaborations
Measure, order and compare objects using familiar metric units of length, area, mass and capacity (VCMMG140)	• recognising the importance of using common units of measurement • recognising and using centimetres and metres, square centimetres, grams and kilograms, and millilitres and litres • measuring the area of rectangles (including squares) by counting the number of square centimetres
Tell time to the minute and investigate the relationship between units of time (VCMMG141)	• recognising there are 60 minutes in an hour and 60 seconds in a minute

Shape	Elaborations
Make models of three-dimensional objects and describe key features (VCMMG142)	• exploring the creation of three-dimensional objects, including prisms and pyramids, using origami

Location and transformation	Elaborations
Create and interpret simple grid maps to show position and pathways (VCMMG143)	• creating a map of the classroom or playground
Identify symmetry in the environment (VCMMG144)	• identifying symmetry in Aboriginal rock carvings or art • identifying symmetry in the natural and built environment
Identify and describe slides and turns found in the natural and built environment (VCMMG145)	• recognising and representing slides and turn used in brickwork around the school • recognising and representing slides and turn used in sporting activities

Geometric reasoning	Elaborations
Identify angles as measures of turn and compare angle sizes in everyday situations (VCMMG146)	• opening doors partially and fully and comparing the size of the angles created • recognising that analogue clocks use the turning of arms to indicate time, and comparing the size of angles between the arms for familiar times

Statistics and Probability

Chance	Elaborations
Conduct chance experiments, identify and describe possible outcomes and recognise variation in	• conducting repeated trials of chance experiments such as tossing a coin or drawing a ball from a bag and identifying the variations between trials

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results (VCMSP147)

Data representation and interpretation	Elaborations
Identify questions or issues for categorical variables. Identify data sources and plan methods of data collection and recording (VCMSP148)	• refining questions and planning investigations that involve collecting data, and carrying out the investigation. For example narrowing the focus of a question such as 'which is the most popular breakfast cereal?' to 'which is the most popular breakfast cereal among students in our class?'
Collect data, organise into categories and create displays using lists, tables, picture graphs and simple column graphs, with and without the use of digital technologies (VCMSP149)	• exploring meaningful and increasingly efficient ways to record data, and representing and reporting the results of investigations • collecting data to investigate features in the natural environment
Interpret and compare data displays (VCMSP150)	• comparing various student-generated data representations and describing their similarities and differences

Level 3 Achievement Standard

Number and Algebra

Students count and order numbers to and from 10 000. They recognise the connection between addition and subtraction, and solve problems using efficient strategies for multiplication with and without the use of digital technology. Students recall addition and multiplication facts for single-digit numbers. They represent money values in various ways and correctly count out change from financial transactions. Students model and represent unit fractions for halves, thirds, quarters, fifths and eighths, and multiples of these up to one. They classify numbers as either odd or even, continue number patterns involving addition or subtraction, and explore simple number sequences based on multiples.

Measurement and Geometry

Students use metric units for length, area, mass and capacity. They tell time to the nearest minute. Students identify symmetry in natural and constructed environments. They use angle size as a measure of turn in real situations and make models of three-dimensional objects. Students match positions on maps with given information and create simple maps.

Statistics and Probability

Students carry out simple data investigations for categorical variables. They interpret and compare data displays. Students conduct chance experiments, list possible outcomes and recognise variations in results.

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Level 4

In Level 4, students extend the number system to simple decimal fractions, and broaden their use of measures and scales.

Students model, represent and order numbers to tens of thousands, and extend place value to tenths and hundredths. They investigate odd and even numbers and explore number patterns based on multiples of 3, 4, 6, 7, 8 and 9. Students develop facility with multiplication facts up to 10 x 10 and related division facts. They investigate simple equivalent fractions and count by halves, thirds and quarters, and locate corresponding elements on a number line. Students use simple decimals to solve money problems including total cost and change. They solve simple number sentences and word problems involving all four operations.

Students use scaled instruments with metric units to measure and compare length, mass, capacity and temperature. They compare shapes and objects using familiar metric units for area and volume, and compare angles with respect to a right angle. Students use 'am' and 'pm' notations, and solve simple time problems, including conversions between units of time. They construct new shapes by combining or splitting common shapes, and create symmetric patterns, pictures and shapes with and without the use of technology. They interpret and use basic maps with simple scales, directions and legends.

Students select and trial different methods for collecting data, including surveys. They construct suitable data displays with and without the use of technology, where there is a many-to-one relationship between elements of graphs and data, and evaluate the effectiveness of different displays. They identify relative likelihood of everyday events, and identify events that are mutually exclusive and events that are independent.

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Level 4 Content Descriptions

Number and Algebra

Number and place value	Elaborations
Investigate and use the properties of odd and even numbers (VCMNA151)	• using the four operations with pairs of odd or even numbers or one odd and one even number, then using the relationships established to check the accuracy of calculations
Recognise, represent and order numbers to at least tens of thousands (VCMNA152)	• reproducing five-digit numbers in words using their numerical representations, and vice versa
Apply place value to partition, rearrange and regroup numbers to at least tens of thousands to assist calculations and solve problems (VCMNA153)	• recognising and demonstrating that the place-value pattern is built on the operations of multiplication or division of tens
Investigate number sequences involving multiples of 3, 4, 6, 7, 8, and 9 (VCMNA154)	• recognising that number sequences can be extended indefinitely, and determining any patterns in the sequences
Recall multiplication facts up to 10 × 10 and related division facts (VCMNA155)	• using known multiplication facts to calculate related division facts • using strategies to recall the multiplication facts • extending multiplication facts (for example 4 by 7 is 28 so 4 by 7 tens is 28 tens)
Develop efficient mental and written strategies and use appropriate digital technologies for multiplication and for division where there is no remainder (VCMNA156)	• using known facts and strategies, such as commutativity, doubling and halving for multiplication, and connecting division to multiplication when there is no remainder

Fractions and decimals	Elaborations
Investigate equivalent fractions used in contexts (VCMNA157)	• exploring the relationship between families of fractions (halves, quarters and eighths or thirds and sixths) by folding a series of paper strips to construct a fraction wall
Count by quarters, halves and thirds, including with mixed numerals. Locate and represent these fractions on a number line (VCMNA158)	• converting mixed numbers to improper fractions and vice versa • investigating the use of fractions and sharing as a way of managing Country: for example taking no more than half the eggs from a nest to protect future bird populations
Recognise that the place value system can be extended to tenths and hundredths. Make connections between fractions and decimal	• using division by 10 to extend the place-value system • using knowledge of fractions to establish equivalences between fractions and decimal notation

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notation (VCMNA159)

Money and financial mathematics	Elaborations
Solve problems involving purchases and the calculation of change to the nearest five cents with and without digital technologies (VCMNA160)	• recognising that not all countries use dollars and cents, eg India uses rupees. • carrying out calculations in another currency as well as in dollars and cents, and identifying both as decimal systems

Patterns and algebra	Elaborations
Explore and describe number patterns resulting from performing multiplication (VCMNA161)	• identifying examples of number patterns in everyday life
Solve word problems by using number sentences involving multiplication or division where there is no remainder (VCMNA162)	• representing a word problem as a number sentence • writing a word problem using a given number sentence
Use equivalent number sentences involving addition and subtraction to find unknown quantities (VCMNA163)	• writing number sentences to represent and answer questions such as: ‘When a number is added to 23 the answer is the same as 57 minus 19. What is the number?’ • using partitioning to find unknown quantities in number sentences
Define a simple class of problems and solve them using an effective algorithm that involves a short sequence of steps and decisions (VCMNA164)	• constructing and applying an algorithm for multiplication of two-digit numbers • partitioning and ordering a set of Australian coins by denomination

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Measurement and Geometry

Using units of measurement	Elaborations
Use scaled instruments to measure and compare lengths, masses, capacities and temperatures (VCMMG165)	• reading and interpreting, to the nearest graduation, the graduated scales on a range of measuring instruments
Compare objects using familiar metric units of area and volume (VCMMG166)	• comparing areas using grid paper • comparing volume using centicubes • recognising that metric units are not the only units used throughout the world, for example measuring the area of floor space using tatami mats (Japan), using squares for room and house area (Australia)
Convert between units of time (VCMMG167)	• identifying and using the correct operation for converting units of time
Use am and pm notation and solve simple time problems (VCMMG168)	• calculating the time spent at school during a normal school day • calculating the time required to travel between two locations • determining arrival time given departure time

Shape	Elaborations
Compare the areas of regular and irregular shapes by informal means (VCMMG169)	• comparing areas using metric units, such as counting the number of square centimetres required to cover two areas by overlaying the areas with a grid of centimetre squares
Compare and describe two dimensional shapes that result from combining and splitting common shapes, with and without the use of digital technologies (VCMMG170)	• identifying common two-dimensional shapes that are part of a composite shape by re-creating it from these shapes • creating two-dimensional shapes from verbal or written instructions
Explain and compare the geometric properties of two-dimensional shapes and three-dimensional objects (VCMMG171)	• describing the similarities and differences between two-dimensional shapes and three-dimensional objects • recognising two-dimensional shapes that are the faces for three-dimensional objects such as prisms, pyramids and platonic solids (including tetrahedrons, cubes and dodecahedrons)

Location and transformation	Elaborations
Use simple scales, legends and directions to interpret information contained in basic maps (VCMMG172)	• identifying the scale used on maps of cities and rural areas in Australia and a city in Indonesia and describing the difference • using directions to find features on a map
Create symmetrical patterns, pictures and shapes with and without digital technologies	• using stimulus materials such as the motifs in Central Asian textiles, Tibetan artefacts, Indian lotus designs and symmetry in Yolngu or Central and Western Desert art

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(VCMMG173)

Geometric reasoning	Elaborations
Compare angles and classify them as equal to, greater than or less than a right angle (VCMMG174)	• creating angles and comparing them to a right angle using digital technologies

Statistics and Probability

Chance	Elaborations
Describe possible everyday events and order their chances of occurring (VCMSP175)	• using lists of events familiar to students and ordering them from 'least likely' to 'most likely' to occur
Identify everyday events where one cannot happen if the other happens (VCMSP176)	• using examples such as weather, which cannot be dry and wet at the same time
Identify events where the chance of one will not be affected by the occurrence of the other (VCMSP177)	• explaining that the probability of a new baby being either a boy or a girl does not depend on the sex of the previous baby

Data representation and interpretation	Elaborations
Select and trial methods for data collection, including survey questions and recording sheets (VCMSP178)	• comparing the effectiveness of different methods of collecting data • choosing the most effective way to collect data for a given investigation
Construct suitable data displays, with and without the use of digital technologies, from given or collected data. Include tables, column graphs and picture graphs where one picture can represent many data values (VCMSP179)	• exploring ways of presenting data and showing the results of investigations • investigating data displays using many-to-one correspondence
Evaluate the effectiveness of different displays in illustrating data features including variability (VCMSP180)	• interpreting data representations in the media and other forums in which symbols represent more than one data value • suggesting questions that can be answered by a given data display and using the display to answer questions

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Level 4 Achievement Standard

Number and Algebra

Students recall multiplication facts to 10 x 10 and related division facts. They choose appropriate strategies for calculations involving multiplication and division, with and without the use of digital technology, and estimate answers accurately enough for the context. Students solve simple purchasing problems with and without the use of digital technology. They locate familiar fractions on a number line, recognise common equivalent fractions in familiar contexts and make connections between fractions and decimal notations up to two decimal places. Students identify unknown quantities in number sentences. They use the properties of odd and even numbers and describe number patterns resulting from multiplication. Students continue number sequences involving multiples of single-digit numbers and unit fractions, and locate them on a number line.

Measurement and Geometry

Students compare areas of regular and irregular shapes, using informal units. They solve problems involving time duration. Students use scaled instruments to measure length, angle, area, mass, capacity and temperature of shapes and objects. They convert between units of time. Students create symmetrical simple and composite shapes and patterns, with and without the use of digital technology. They classify angles in relation to a right angle. Students interpret information contained in maps.

Statistics and Probability

Students describe different methods for data collection and representation, and evaluate their effectiveness. They construct data displays from given or collected data, with and without the use of digital technology. Students list the probabilities of everyday events. They identify dependent and independent events.

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Level 5

In Level 5, students extend decimal fractions to thousandths, and explore the ideas of factors, multiples and divisibility.

Students use estimation and rounding for all four operations, with and without the use of technology for calculation. They solve multiple digit problems involving addition, subtraction, multiplication and division by single digit divisors with remainders. Students represent, compare and order unit fractions, and decimal fractions, and represent them on a number line. They construct simple budgets for familiar events and activities. They solve numbers sentences involving division, and create number patterns involving fractions and decimals.

Students choose and use suitable metric and other units for measurement of length, angle, area, volume, capacity and mass. They calculate the perimeter and area of rectangles, and construct specified angles using protractors and other relevant technologies. Students use 12 and 24 hour time systems, with measurements and conversions to seconds. They use grid reference systems to describe location and connect three-dimension objects with two-dimensional representations. They translate, reflect and rotate shapes with and without the use of technology, and identify point and line symmetries. They explore similarity of familiar shapes through enlargement.

Students pose questions to collect categorical and numerical data by observation and survey, and represent the data in a variety of ways with and without the use of technology. They describe and interpret data sets in context. Students recognise that probabilities are measured on a scale of 0 to 1 (inclusive), and represent the probability of events from simple experiments using fractions.

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Level 5 Content Descriptions

Number and Algebra

Number and place value	Elaborations
Identify and describe factors and multiples of whole numbers and use them to solve problems (VCMNA181)	• exploring factors and multiples using number sequences • using simple divisibility tests
Use estimation and rounding to check the reasonableness of answers to calculations (VCMNA182)	• recognising the usefulness of estimation to check calculations • applying mental strategies to estimate the result of calculations, such as estimating the cost of a supermarket trolley load
Solve problems involving multiplication of large numbers by one- or two-digit numbers using efficient mental, written strategies and appropriate digital technologies (VCMNA183)	• using rounding and making estimates for computations • exploring techniques for multiplication such as the area model, the Italian lattice method or the partitioning of numbers • applying the distributive law and using arrays to model multiplication and explain calculation strategies
Solve problems involving division by a one digit number, including those that result in a remainder (VCMNA184)	• using the fact that equivalent division calculations result if both numbers are divided by the same factor • using rounding and making estimates for computations • interpreting and representing the remainder in division calculations sensibly for the context
Use efficient mental and written strategies and apply appropriate digital technologies to solve problems (VCMNA185)	• choosing between mental, written and a technology-based computation depending on the nature of the problems and the purpose for computation • using technology to solve problems and check the reasonableness of answers
Recognise, represent and order numbers to at least hundreds of thousands (VCMNA186)	• reproducing six-digit numbers in words using their numerical representations, and vice versa

Fractions and decimals	Elaborations
Compare and order common unit fractions and locate and represent them on a number line (VCMNA187)	• recognising the connection between the order of unit fractions and their denominators
Investigate strategies to solve problems involving addition and subtraction of fractions with the same denominator (VCMNA188)	• modelling and solving addition and subtraction problems involving fractions by using jumps on a number line, or making diagrams of fractions as parts of shapes
Recognise that the place value system can be extended beyond	• using knowledge of place value and division by 10 to extend the number system to thousandths and beyond

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hundredths (VCMNA189)	• recognising the equivalence of one thousandths and 0.001
Compare, order and represent decimals (VCMNA190)	• locating decimals on a number line

Money and financial mathematics	Elaborations
Create simple financial plans (VCMNA191)	• creating a simple budget for a class fundraising event • identifying the GST component of invoices and receipts

Patterns and algebra	Elaborations
Describe, continue and create patterns with fractions, decimals and whole numbers resulting from addition and subtraction (VCMNA192)	• using the number line or diagrams to create patterns involving fractions or decimals
Use equivalent number sentences involving multiplication and division to find unknown quantities (VCMNA193)	• using relevant problems to develop number sentences
Follow a mathematical algorithm involving branching and repetition (iteration) (VCMNA194)	• simulating a simple random walk • manipulating sets of numbers using a given rule, for example, if a number is even halve it; if a number is odd, subtract 1 then halve it

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Measurement and Geometry

Using units of measurement	Elaborations
Choose appropriate units of measurement for length, area, volume, capacity and mass (VCMMG195)	• investigating alternative measures of scale to demonstrate that these vary between countries and change over time. For example, temperature measurement in Australia, Indonesia, Japan and USA • recognising that some units of measurement are better suited for some tasks than others, for example kilometres rather than metres to measure the distance between two towns
Calculate the perimeter and area of rectangles and the volume and capacity of prisms using familiar metric units (VCMMG196)	• exploring efficient ways of calculating the perimeters of rectangles such as adding the length and width together and doubling the result • exploring efficient ways of finding the areas of rectangles • measuring volume and capacity by counting the number of cubes (cubic centimetres) • exploring efficient ways of finding the volume and capacity of rectangular prisms and cubes
Compare 12- and 24-hour time systems and convert between them (VCMMG197)	• investigating the ways time was and is measured in different Aboriginal Country, such as using tidal change • using units hours, minutes and seconds

Shape	Elaborations
Connect three-dimensional objects with their nets and other two-dimensional representations (VCMMG198)	• identifying the shape and relative position of each face of a solid to determine the net of the solid, including that of prisms and pyramids • representing two-dimensional shapes such as photographs, sketches and images created by digital technologies

Location and transformation	Elaborations
Use a grid reference system to describe locations. Describe routes using landmarks and directional language (VCMMG199)	• comparing aerial views of Country, desert paintings and maps with grid references • creating a grid reference system for the classroom and using it to locate objects and describe routes from one object to another
Describe translations, reflections and rotations of two-dimensional shapes. Identify line and rotational symmetries (VCMMG200)	• identifying and describing the line and rotational symmetry of a range of two-dimensional shapes, by manually cutting, folding and turning shapes and by using digital technologies • identifying the effects of transformations by manually flipping, sliding and turning two-dimensional shapes and by using digital technologies
Apply the enlargement transformation to familiar two dimensional shapes and explore the properties of the resulting image compared with the original (VCMMG201)	• using digital technologies to enlarge shapes • using a grid system to enlarge a favourite image or cartoon

Geometric reasoning	Elaborations

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Estimate, measure and compare angles using degrees. Construct angles using a protractor (VCMMG202)	• measuring and constructing angles using both 180° and 360° protractors • recognising that angles have arms and a vertex, and that size is the amount of turn required for one arm to coincide with the other

Statistics and Probability

Chance	Elaborations
List outcomes of chance experiments involving equally likely outcomes and represent probabilities of those outcomes using fractions (VCMSP203)	• commenting on the likelihood of winning simple games of chance by considering the number of possible outcomes and the consequent chance of winning in simple games of chance such as jan-ken-pon (also known as rock-paper-scissors)
Recognise that probabilities range from 0 to 1 (VCMSP204)	• investigating the probabilities of all outcomes for a simple chance experiment and verifying that their sum equals 1

Data representation and interpretation	Elaborations
Pose questions and collect categorical or numerical data by observation or survey (VCMSP205)	• posing questions about insect diversity in the playground, collecting data by taping a one-metre-square piece of paper to the playground and observing the type and number of insects on it over time
Construct displays, including column graphs, dot plots and tables, appropriate for data type, with and without the use of digital technologies (VCMSP206)	• identifying the best methods of presenting data to illustrate the results of investigations and justifying the choice of representations
Describe and interpret different data sets in context (VCMSP207)	• using and comparing data representations for different data sets to help decision making

Level 5 Achievement Standard

Number and Algebra

Students solve simple problems involving the four operations using a range of strategies including digital technology. They estimate to check the reasonableness of answers and approximate answers by rounding. Students identify and describe factors and multiples. They explain plans for simple budgets. Students order decimals and unit fractions and locate them on a number line. Students add and subtract fractions with the same denominator. They find unknown quantities in number sentences and continue patterns by adding or subtracting fractions and decimals.

Measurement and Geometry

Students use appropriate units of measurement for length, area, volume, capacity and mass, and calculate perimeter and area of rectangles and volume, and capacity of rectangular prisms. They convert between 12 and 24-hour time. Students use a grid reference system to locate landmarks. They estimate angles, and use protractors and digital technology to construct and measure

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angles. Students connect three-dimensional objects with their two-dimensional representations. They describe transformations of two-dimensional shapes and identify line and rotational symmetry.

Statistics and Probability

Students pose questions to gather data and construct various displays appropriate for the data, with and without the use of digital technology. They compare and interpret different data sets. Students list outcomes of chance experiments with equally likely outcomes and assign probabilities as a number from 0 to 1.

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Level 6

In Level 6, students work with prime, composite, square and triangular numbers and carry out mental, written and technology based computation to solve whole number problems involving all four operations. They explore everyday situations involving integers, and use a number line to represent them. They scale decimals by powers of ten, and add and subtract decimals with and without technology, and estimate their answers. Students calculate simple percentage discounts, multiply decimals by whole numbers, carry out divisions with terminating decimal remainders, and use simple fraction, decimal and percentage equivalences with and without technology. They create sequences involving whole numbers, fractions and decimals, describe their rules, and use brackets and order of operations to write number sentences involving multiple operations.

Students use decimals for metric measurement, convert between units, recognise the prefixes used in metric measurements, and relate and compare measures and units, including capacity and volume. They develop and use timetables. Students investigate combinations of transformations with and without technology, and use the Cartesian coordinate system to describe location in the plane. They investigate the sum of angles at a point on a line and vertically opposite angles.

Students carry out experiments involving chance with and without technology, compare variation in frequencies across experiments with expected frequencies, and use fractions, decimals and percentages to describe probabilities. They interpret a range of data displays, including those for two categorical variables, and interpret data presented in the media.

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Level 6 Content Descriptions

Number and Algebra

Number and place value	Elaborations
Identify and describe properties of prime, composite, square and triangular numbers (VCMNA208)	• understanding that some numbers have special properties and that these properties can be used to solve problems • representing composite numbers as a product of their prime factors and using this form to simplify calculations by cancelling common primes • understanding that if a number is divisible by a composite number then it is also divisible by the prime factors of that number.
Select and apply efficient mental and written strategies and appropriate digital technologies to solve problems involving all four operations with whole numbers and make estimates for these computations (VCMNA209)	• applying strategies already developed for solving problems involving small numbers to those involving large numbers • applying a range of strategies to solve realistic problems and commenting on the efficiency of different strategies • forming simple single digit estimates with consideration of order of magnitude of the result
Investigate everyday situations that use integers. Locate and represent these numbers on a number line (VCMNA210)	• recognising that integers form an ordered infinite set {…-3, -2, -1, 0, 1, 2, 3…} with no first element or last element • solving everyday additive problems using a number line • investigating everyday situations that use integers, such as temperatures • using number lines to position and order integers around zero

Fractions and decimals	Elaborations
Compare fractions with related denominators and locate and represent them on a number line (VCMNA211)	• demonstrating equivalence between fractions using drawings and models
Solve problems involving addition and subtraction of fractions with the same or related denominators (VCMNA212)	• understanding the processes for adding and subtracting fractions with related denominators and fractions as an operator, in preparation for calculating with all fractions • solving realistic additive (addition and subtraction) problems involving fractions to develop understanding of equivalent fractions and the use of fractions as operators • modelling and solving additive problems involving fractions by using methods such as jumps on a number line, or by making diagrams of fractions as parts of shapes
Find a simple fraction of a quantity where the result is a whole number, with and without digital technologies (VCMNA213)	• recognising that finding one third of a quantity is the same as dividing by 3
Add and subtract decimals, with and without digital technologies, and use estimation and rounding to check the reasonableness of answers	• extending whole-number strategies to explore and develop meaningful written strategies for addition and subtraction of decimal numbers to thousandths • exploring and practising efficient methods for solving problems requiring operations on decimals, to gain fluency with calculating with decimals and with recognising appropriate operations

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(VCMNA214)	
Multiply decimals by whole numbers and perform divisions by non-zero whole numbers where the results are terminating decimals, with and without digital technologies (VCMNA215)	• interpreting the results of calculations to provide an answer appropriate to the context
Multiply and divide decimals by powers of 10 (VCMNA216)	• multiplying and dividing decimals by multiples of powers of 10
Make connections between equivalent fractions, decimals and percentages (VCMNA217)	• connecting fractions, decimals and percentages as different representations of the same number, moving fluently between representations and choosing the appropriate one for the problem being solved

Money and financial mathematics	Elaborations
Investigate and calculate percentage discounts of 10%, 25% and 50% on sale items, with and without digital technologies (VCMNA218)	• using authentic information to calculate prices on sale goods

Patterns and algebra	Elaborations
Continue and create sequences involving whole numbers, fractions and decimals. Describe the rule used to create the sequence (VCMNA219)	• identifying and generalising number patterns • investigating additive and multiplicative patterns such as the number of tiles in a geometric pattern, or the number of dots or other shapes in successive repeats of a strip or border pattern looking for patterns in the way the numbers increase/decrease
Explore the use of brackets and order of operations to write number sentences (VCMNA220)	• appreciating the need for rules to complete multiple operations within the same number sentence
Design algorithms involving branching and iteration to solve specific classes of mathematical problems (VCMNA221)	• implementing algorithms such as the Euclidean division algorithm • devising flowcharts to represent algorithms for a common processes such as adding two fractions

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Measurement and Geometry

Using units of measurement	Elaborations
Connect decimal representations to the metric system (VCMMG222)	• recognising the equivalence of measurements such as 1.25 metres and 125 centimetres
Convert between common metric units of length, mass and capacity (VCMMG223)	• identifying and using the correct operations when converting units including millimetres, centimetres, metres, kilometres, milligrams, grams, kilograms, tonnes, millilitres, litres, kilolitres and megalitres • recognising the significance of the prefixes in units of measurement
Solve problems involving the comparison of lengths and areas using appropriate units (VCMMG224)	• recognising and investigating familiar objects using concrete materials and digital technologies
Connect volume and capacity and their units of measurement (VCMMG225)	• recognising that 1ml is equivalent to $1cm^3$
Interpret and use timetables (VCMMG226)	• planning a trip involving one or more modes of public transport • developing a timetable of daily activities
Measure, calculate and compare elapsed time (VCMMG227)	• recognising that some units of time are better suited for some tasks than others, for example, seconds rather than minutes to measure the duration of a short event • using a stopwatch to measure and compare the duration of a series of events, for example, how long does it take to jump a skipping rope 100 times • using the starting and finishing times to calculate and compare elapsed time, for example, the time taken to walk to school compared to the time taken to ride to school

Shape	Elaborations
Construct simple prisms and pyramids (VCMMG228)	• considering the history and significance of pyramids from a range of cultural perspectives including those structures found in China, Korea and Indonesia • constructing prisms and pyramids from nets, and skeletal models

Location and transformation	Elaborations
Investigate the effect of combinations of transformations on simple and composite shapes, including creating tessellations, with and without the use of digital technologies (VCMMG229)	• understanding which attributes of a shape remain the same or change under the effects of transformations such as size, shape or orientation • creating tessellations with simple shapes or simple composite shapes
Introduce the Cartesian coordinate system using all four quadrants	• understanding that the Cartesian plane provides a graphical or visual way of describing location

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(VCMMG230)

Geometric reasoning	Elaborations
Investigate, with and without digital technologies, angles on a straight line, angles at a point and vertically opposite angles. Use results to find unknown angles (VCMMG231)	• identifying the size of a right angle as 90° and defining acute, obtuse, straight and reflex angles • measuring, estimating and comparing angles in degrees and classifying angles according to their sizes • investigating the use of rotation and symmetry in the diagrammatic representations of kinship relationships of Central and Western Desert people • recognising and using the two alternate conventions for naming angles

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Statistics and Probability

Chance	Elaborations
Describe probabilities using fractions, decimals and percentages (VCMSP232)	• investigating games of chance popular in different cultures
Conduct chance experiments with both small and large numbers of trials using appropriate digital technologies (VCMSP233)	• conducting repeated trials of chance experiments, identifying the variation between trials and realising that the results tend to the prediction with larger numbers of trials
Compare observed frequencies across experiments with expected frequencies (VCMSP234)	• predicting likely outcomes from a run of chance events and distinguishing these from surprising results

Data representation and interpretation	Elaborations
Construct, interpret and compare a range of data displays, including side-by-side column graphs for two categorical variables (VCMSP235)	• comparing different student-generated diagrams, tables and graphs, describing their similarities and differences and commenting on the usefulness of each representation for interpreting the data • understanding that data can be represented in different ways, sometimes with one symbol representing more than one piece of data, and that it is important to read all information about a representation before making judgments
Interpret secondary data presented in digital media and elsewhere (VCMSP236)	• investigating data representations in the media and discussing what they illustrate and the messages the people who created them might want to convey • identifying potentially misleading data representations in the media, such as graphs with broken axes or non-linear scales, graphics not drawn to scale, data not related to the population about which the claims are made, and pie charts in which the whole pie does not represent the entire population about which the claims are made
Pose and refine questions to collect categorical or numerical data by observation or survey (VCMSP237)	• narrowing or broadening a question to be posed for an observation or survey, for example, 'which of these ice cream flavours – vanilla or chocolate?' to 'do you like ice cream, if yes what is your favourite flavour?'

Level 6 Achievement Standard

Number and Algebra

Students recognise the properties of prime, composite, square and triangular numbers and determine sets of these numbers. They solve problems that involve all four operations with whole numbers and describe the use of integers in everyday contexts. Students locate fractions and integers on a number line and connect fractions, decimals and percentages as different representations of the same number. They solve problems involving the addition and subtraction of related fractions. Students calculate a simple fraction of a quantity and calculate common percentage discounts on sale items, with and without the use of digital technology. They make connections between the powers of 10 and the multiplication and division of decimals. Students add, subtract and multiply decimals and divide decimals where the result is rational. Students write number sentences using brackets and order of operations, and specify rules used to generate sequences involving whole numbers, fractions and decimals. They use ordered pairs of integers to represent coordinates of points and locate a point in any one of the four quadrants on the Cartesian plane.

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Measurement and Geometry

Students relate decimals to the metric system and choose appropriate units of measurement to perform a calculation. They solve problems involving time, length and area, and make connections between capacity and volume. Students interpret a variety of everyday timetables. They solve problems using the properties of angles and investigate simple combinations of transformations in the plane, with and without the use of digital technology. Students construct simple prisms and pyramids.

Statistics and Probability

Students interpret and compare a variety of data displays, including displays for two categorical variables. They analyse and evaluate data from secondary sources. Students compare observed and expected frequencies of events, including those where outcomes of trials are generated with the use of digital technology. They specify, list and communicate probabilities of events using simple ratios, fractions, decimals and percentages.

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Level 7

In Level 7, students work with powers of whole numbers, use index notation, represent numbers as products of powers of prime numbers, and investigate square roots of perfect squares. They use number properties to assist with calculation and order, and to add and subtract integers. Students find equivalent fractions, represent positive and negative fractions and mixed numbers on a number line and add, subtract, multiply and divide fractions and decimals with and without the use of technology. They express one quantity as a fraction of another, round to a specified number of decimal places, and convert between fractions, decimals and percentages. They find percentages of quantities and one quantity as a percentage of another. They solve simple ratio problems and calculate best buys with and without the use of technology.

Students use variables to express relationships in real life data, and interpret and analyse corresponding graphs. They use pro-numerals to construct simple algebraic expressions and substitute numerical values into these. They solve simple linear equations and plot points on the Cartesian plane.

Students use formulas for calculating areas of triangles, rectangles and related shapes, and volumes of cubes and rectangular prisms. They form two-dimensional representations of prisms, buildings and other structures. They use simple combinations of transformations, with and without technology, to create geometric patterns and identify line and point symmetry, apply parallel line and transversal angle properties, angles sums in triangles and quadrilaterals, classify triangles and quadrilaterals, and construct them using compass and straight edge and dynamic geometry technology.

Students construct sample spaces for simple experiments involving chance, and assign probabilities to outcomes. They use data from primary and secondary sources to investigate issues of interest, and employ data displays such as dot plots and stem and leaf plots to compare data sets, and calculate measures of centre and simple measures of spread to analyse and interpret the data.

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Level 7 Content Descriptions

Number and Algebra

Number and place value	Elaborations
Investigate index notation and represent whole numbers as products of powers of prime numbers (VCMNA238)	• defining and comparing prime and composite numbers and explaining the difference between them • applying knowledge of factors to strategies for expressing whole numbers as products of powers of prime factors, such as repeated division by prime factors or creating factor trees • solving problems involving lowest common multiples and greatest common divisors (highest common factors) for pairs of whole numbers by comparing their prime factorisation
Investigate and use square roots of perfect square numbers (VCMNA239)	• investigating square numbers such as 25 and 36 and developing square-root notation • investigating between which two whole numbers a square root lies
Apply the associative, commutative and distributive laws to aid mental and written computation and make estimates for these computations (VCMNA240)	• simplifying calculations • forming simple estimates for calculations involving multiple and/or combined operations
Compare, order, add and subtract integers (VCMNA241)	• using a variety of models to represent, add and subtract integers

Real numbers	Elaborations
Compare fractions using equivalence. Locate and represent positive and negative fractions and mixed numbers on a number line (VCMNA242)	• exploring equivalence among families of fractions by using a fraction wall or a number line (for example by using a fraction wall to show that 2/3 is the same as 4/6 and 6/9)
Solve problems involving addition and subtraction of fractions, including those with unrelated denominators (VCMNA243)	• exploring and developing efficient strategies to solve additive problems involving fractions (for example by using fraction walls or rectangular arrays with dimensions equal to the denominators)
Multiply and divide fractions and decimals using efficient written strategies and digital technologies (VCMNA244)	• investigating multiplication of fractions and decimals, using strategies including patterning and multiplication as repeated addition, with both concrete materials and digital technologies, and identifying the processes for division as the inverse of multiplication
Express one quantity as a fraction of another, with and without the use of digital technologies (VCMNA245)	• using examples for the quantities to be expressed and understanding the reasons for the calculations
Round decimals to a specified	• using rounding to estimate the results of calculations with whole numbers and

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number of decimal places (VCMNA246)	decimals, and understanding the conventions for rounding
Connect fractions, decimals and percentages and carry out simple conversions (VCMNA247)	• justifying choices of written, mental or calculator strategies for solving specific problems including those involving large numbers • understanding that quantities can be represented by different number types and calculated using various operations, and that choices need to be made about each • calculating the percentage of the total local municipal area set aside for parkland, manufacturing, retail and residential dwellings to compare land use
Find percentages of quantities and express one quantity as a percentage of another, with and without digital technologies. (VCMNA248)	• using authentic problems to express quantities as percentages of other amounts
Recognise and solve problems involving simple ratios (VCMNA249)	• understanding that rate and ratio problems can be solved using fractions or percentages and choosing the most efficient form to solve a particular problem

Money and financial mathematics	Elaborations
Investigate and calculate 'best buys', with and without digital technologies (VCMNA250)	• applying the unitary method to identify ‘best buys’ situations, such as comparing the cost per 100g

Patterns and algebra	Elaborations
Introduce the concept of variables as a way of representing numbers using letters (VCMNA251)	• understanding that arithmetic laws are powerful ways of describing and simplifying calculations and that using these laws leads to the generality of algebra
Create algebraic expressions and evaluate them by substituting a given value for each variable (VCMNA252)	• using authentic formulas to perform substitutions
Extend and apply the laws and properties of arithmetic to algebraic terms and expressions (VCMNA253)	• identifying order of operations in contextualised problems, preserving the order by inserting brackets in numerical expressions, then recognising how order is preserved by convention • moving fluently between algebraic and word representations as descriptions of the same situation
Design and implement mathematical algorithms using a simple general purpose programming language (VCMNA254)	• finding the sum of a set of consecutive numbers using a loop structure • constructing geometric patterns such as a honeycomb, using dynamic geometry functionality

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Linear and non-linear relationships	Elaborations
Given coordinates, plot points on the Cartesian plane, and find coordinates for a given point (VCMNA255)	• plotting points from a table of integer values and recognising simple patterns, such as points that lie on a straight line
Solve simple linear equations (VCMNA256)	• solving equations using concrete materials, such as the balance model, and explain the need to do the same thing to each side of the equation using substitution to check solutions • investigating a range of strategies to solve equations
Investigate, interpret and analyse graphs from real life data, including consideration of domain and range (VCMNA257)	• using travel graphs to investigate and compare the distance travelled to and from school • interpreting features of travel graphs such as the slope of lines and the meaning of horizontal lines • using graphs of evaporation rates to explore water storage • describing and comparing temperature during a day at different times of the year from the corresponding graphs

Measurement and Geometry

Using units of measurement	Elaborations
Establish the formulas for areas of rectangles, triangles and parallelograms and use these in problem solving (VCMMG258)	• building on the understanding of the area of rectangles to develop formulas for the area of triangles • establishing that the area of a triangle is half the area of an appropriate rectangle • using area formulas for rectangles and triangles to solve problems involving areas of surfaces
Calculate volumes of rectangular prisms (VCMMG259)	• investigating volumes of cubes and rectangular prisms and establishing and using the formula $V = l \times b \times h$ • understanding and using cubic units when interpreting and finding volumes of cubes and rectangular prisms

Shape	Elaborations
Draw different views of prisms and solids formed from combinations of prisms (VCMMG260)	• using aerial views of buildings and other 3-D structures to visualise the structure of the building or prism

Location and transformation	Elaborations
Describe translations, reflections in an axis, and rotations of multiples of 90° on the Cartesian plane using coordinates. Identify line and rotational symmetries (VCMMG261)	• describing patterns and investigating different ways to produce the same transformation such as using two successive reflections to provide the same result as a translation • creating and re-creating patterns using combinations of reflections and rotations, using digital technologies

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Geometric reasoning	Elaborations
Classify triangles according to their side and angle properties and describe quadrilaterals (VCMMG262)	• identifying side and angle properties of scalene, isosceles, right-angled and obtuse-angled triangles • describing squares, rectangles, rhombuses, parallelograms, kites and trapeziums
Demonstrate that the angle sum of a triangle is 180° and use this to find the angle sum of a quadrilateral (VCMMG263)	• using concrete materials and digital technologies to investigate the angle sum of a triangle and quadrilateral
Identify corresponding, alternate and co-interior angles when two straight lines are crossed by a transversal (VCMMG264)	• defining and classifying pairs of angles as complementary, supplementary, adjacent and vertically opposite
Investigate conditions for two lines to be parallel and solve simple numerical problems using reasoning (VCMMG265)	• constructing parallel and perpendicular lines using their properties, a pair of compasses and a ruler, and dynamic geometry software • defining and identifying the relationships between alternate, corresponding and co-interior angles for a pair of parallel lines cut by a transversal

Statistics and Probability

Chance	Elaborations
Construct sample spaces for single-step experiments with equally likely outcomes (VCMSP266)	• discussing the meaning of probability terminology. For example, probability, sample space, favourable outcomes, trial, events and experiments • distinguishing between equally likely outcomes and outcomes that are not equally likely
Assign probabilities to the outcomes of events and determine probabilities for events (VCMSP267)	• expressing probabilities as decimals, fractions and percentages

Data representation and interpretation	Elaborations
Identify and investigate issues involving numerical data collected from primary and secondary sources (VCMSP268)	• obtaining secondary data from newspapers, the Internet and the Australian Bureau of Statistics • investigating secondary data relating to the distribution and use of non-renewable resources around the world
Construct and compare a range of data displays including stem-and-leaf plots and dot plots (VCMSP269)	• understanding that some data representations are more appropriate than others for particular data sets, and answering questions about those data sets • using ordered stem-and-leaf plots to record and display numerical data collected in a class investigation, such as constructing a class plot of height in centimetres on a shared stem-and-leaf plot for which the stems 12, 13, 14, 15, 16 and 17 have been produced
Calculate mean, median, mode and	• understanding that summarising data by calculating measures of centre and

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range for sets of data. Interpret these statistics in the context of data (VCMSP270)	spread can help make sense of the data
Describe and interpret data displays using median, mean and range (VCMSP271)	• using mean and median to compare data sets and explaining how outliers may affect the comparison • locating mean, median and range on graphs and connecting them to real life

Level 7 Achievement Standard

Number and Algebra

Students solve problems involving the order, addition and subtraction of integers. They make the connections between whole numbers and index notation and the relationship between perfect squares and square roots. They solve problems involving all four operations with fractions, decimals, percentages and their equivalences, and express fractions in their simplest form. Students compare the cost of items to make financial decisions, with and without the use of digital technology. They make simple estimates to judge the reasonableness of results. Students use variables to represent arbitrary numbers and connect the laws and properties of number to algebra and substitute numbers into algebraic expressions. They assign ordered pairs to given points on the Cartesian plane and interpret and analyse graphs of relations from real data. Students develop simple linear models for situations, make predictions based on these models, solve related equations and check their solutions.

Measurement and Geometry

Students use formulas for the area and perimeter of rectangles. They classify triangles and quadrilaterals and represent transformations of these shapes on the Cartesian plane, with and without the use of digital technology. Students name the types of angles formed by transversals crossing parallel lines and solve simple numerical problems involving these lines and angles. They describe different views of three-dimensional objects, and use models, sketches and digital technology to represent these views. Students calculate volumes of rectangular prisms.

Statistics and Probability

Students identify issues involving the collection of discrete and continuous data from primary and secondary sources. They construct stem-and-leaf plots and dot-plots. Students identify or calculate mean, mode, median and range for data sets, using digital technology for larger data sets. They describe the relationship between the median and mean in data displays. Students determine the sample space for simple experiments with equally likely outcomes, and assign probabilities outcomes.

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Level 8

In Level 8, students consolidate their proficiency with the four arithmetic operations, and combinations of these, for general computation involving natural numbers, integers and rational numbers, with and without the use of technology. They represent these numbers on the real number line. They extend the use of indices and develop the index laws using number examples. Students investigate the relationship between decimal and fraction representations of rational numbers (terminating and recurring decimals) and work with some irrational real numbers such as square roots and multiples and fractions of π *(pi)*. They solve a range of problems involving ratios, proportions, percentages and rates, with and without the use of digital technologies.

Students generalise from number to algebra, and expand, factorise, simplify and substitute into simple algebraic expressions. They plot linear relations on the Cartesian plane, with and without the use of digital technology, solve linear equations and apply linear models.

Students convert between units for area and for volume, and solve problems involving duration using 12-hour and 24-hour time, within a given time zone. They develop and use formulas for calculating perimeters and areas of quadrilaterals and circles, and volumes of prisms, and solve related measurement problems.

Students use congruence and transformations to establish properties of plane shapes related to sides, angles and symmetry, and solve related problems.

Students use the logical connectives ‘not’, ‘and’, ‘or’ and ‘either … or’ to relate events to probabilities, and use Venn diagrams and two-way tables to calculate probabilities. They develop an understanding that probabilities range from 0 to 1 and that the sum of probabilities for events in a sample space is 1.

Students investigate and use various techniques for collecting data, including random sampling. They use digital technology to explore the variability of proportions and means in random samples drawn from a given population, and investigate the effect of individual data values, including outliers, on the measure of centre (average).

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Level 8 Content Descriptions

Number and Algebra

Number and place value	Elaborations
Use index notation with numbers to establish the index laws with positive integral indices and the zero index (VCMNA272)	• evaluating numbers expressed as powers of positive integers
Carry out the four operations with rational numbers and integers, using efficient mental and written strategies and appropriate digital technologies and make estimates for these computations (VCMNA273)	• using patterns to assist in finding rules for the multiplication and division of integers • using the number line to develop strategies for adding and subtracting rational numbers • making an estimate for the total of a family weekly grocery bill with consideration of accuracy of the estimate, or for problems involving the circumference and area of a circle

Real numbers	Elaborations
Investigate terminating and recurring decimals (VCMNA274)	• recognising terminating, recurring and non-terminating decimals and choosing their appropriate representations
Investigate the concept of irrational numbers, including π (VCMNA275)	• understanding that the real number system includes irrational numbers
Solve problems involving the use of percentages, including percentage increases and decreases and percentage error, with and without digital technologies (VCMNA276)	• using percentages to solve problems, including those involving mark-ups, discounts, and GST • using percentages to calculate population increases and decreases • using percentage error to compare relative size of error in calculations involving a given or actual value, and an estimated or measured value
Solve a range of problems involving rates and ratios, including distance-time problems for travel at a constant speed, with and without digital technologies (VCMNA277)	• understanding that rate and ratio problems can be solved using fractions or percentages and choosing the most efficient form to solve a particular problem • calculating population growth rates in Australia and Asia and explaining their difference • finding one of distance travelled, time taken or average speed given the other two quantities

Money and financial mathematics	Elaborations
Solve problems involving profit and loss, with and without digital technologies (VCMNA278)	• expressing profit and loss as a percentage of cost or selling price, comparing the difference • investigating the methods used in retail stores to express discounts

Patterns and algebra	Elaborations
Extend and apply the distributive law to the expansion of algebraic	• applying the distributive law to the expansion of algebraic expressions using strategies such as the area model

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expressions (VCMNA279)	
Factorise algebraic expressions by identifying numerical factors (VCMNA280)	• recognising the relationship between factorising and expanding • identifying the greatest common divisor (highest common factor) of numeric and algebraic expressions and using a range of strategies to factorise algebraic expressions
Simplify algebraic expressions involving the four operations (VCMNA281)	• understanding that the laws used with numbers can also be used with algebra
Use algorithms and related testing procedures to identify and correct errors (VCMNA282)	• debugging search and sort programs • testing a number for divisibility

Linear and non-linear relationships	Elaborations
Plot linear relationships on the Cartesian plane with and without the use of digital technologies (VCMNA283)	• completing a table of values, plotting the resulting points and determining whether the relationship is linear • finding the rule for a linear relationship
Solve linear equations using algebraic and graphical techniques. Verify solutions by substitution (VCMNA284)	• solving real life problems by using variables to represent unknowns
Plot graphs of non-linear real life data with and without the use of digital technologies, and interpret and analyse these graphs (VCMNA285)	• investigating different combinations of length and width of a rectangle for a fixed area, and drawing the corresponding graph • using graphs to analysing change in the value of a currency against another currency over a specified period

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Measurement and Geometry

Using units of measurement	Elaborations
Choose appropriate units of measurement for area and volume and convert from one unit to another (VCMMG286)	• choosing units for area including mm^2, cm^2, m^2, hectares, km^2, and units for volume including mm^3, cm^3, m^3 • recognising that the conversion factors for area units are the squares of those for the corresponding linear units • recognising that the conversion factors for volume units are the cubes of those for the corresponding linear units
Find perimeters and areas of parallelograms, trapeziums, rhombuses and kites (VCMMG287)	• establishing and using formulas for areas such as trapeziums, rhombuses and kites
Investigate the relationship between features of circles such as circumference, area, radius and diameter. Use formulas to solve problems involving determining radius, diameter, circumference and area from each other (VCMMG288)	• investigating the circumference and area of circles with materials or by measuring, to establish an understanding of formulas • investigating the area of circles using a square grid or by rearranging a circle divided into sectors • solving problems given one of radius, diameter, circumference or area of a circle, then the other quantities are determined from this
Develop the formulas for volumes of rectangular and triangular prisms and prisms in general. Use formulas to solve problems involving volume (VCMMG289)	• investigating the relationship between volumes of rectangular and triangular prisms
Solve problems involving duration, including using 12- and 24-hour time within a single time zone (VCMMG290)	• identifying regions in Australia and countries in Asia that are in the same time zone

Geometric reasoning	Elaborations
Define congruence of plane shapes using transformations and use transformations of congruent shapes to produce regular patterns in the plane including tessellations with and without the use of digital technology (VCMMG291)	• understanding the properties that determine congruence of triangles and recognising which transformations create congruent figures • establishing that two figures are congruent if one shape lies exactly on top of the other after one or more transformations (translation, reflection, rotation), and recognising that the matching sides and the matching angles are equal • exploring tiling patterns in art and design
Develop the conditions for congruence of triangles (VCMMG292)	• investigating the minimal conditions needed for the unique construction of triangles, leading to the establishment of the conditions for congruence (SSS, SAS, ASA and RHS) • solving problems using the properties of congruent figures • constructing triangles using the conditions for congruence
Establish properties of quadrilaterals using congruent triangles and angle	• establishing the properties of squares, rectangles, parallelograms, rhombuses, trapeziums and kites

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properties, and solve related numerical problems using reasoning (VCMMG293)	• identifying properties related to side lengths, parallel sides, angles, diagonals and symmetry

Statistics and Probability

Chance	Elaborations
Identify complementary events and use the sum of probabilities to solve problems (VCMSP294)	• identifying the complement of familiar events • understanding that probabilities range between 0 to 1 and that calculating the probability of an event allows the probability of its complement to be found
Describe events using language of 'at least', exclusive 'or' (A or B but not both), inclusive 'or' (A or B or both) and 'and' (VCMSP295)	• posing 'and', 'or' and 'not' probability questions about objects or people
Represent events in two-way tables and Venn diagrams and solve related problems (VCMSP296)	• using Venn diagrams and two-way tables to calculate probabilities for events, satisfying 'and', 'or' and 'not' conditions • understanding that representing data in Venn diagrams or two-way tables facilitates the calculation of probabilities • collecting data to answer the questions using Venn diagrams or two-way tables

Data representation and interpretation	Elaborations
Distinguish between a population and a sample and investigate techniques for collecting data, including census, sampling and observation (VCMSP297)	• identifying situations where data can be collected by census and those where a sample is appropriate • investigating the differences between convenience, judgemental and simple random sampling from a population
Explore the practicalities and implications of obtaining data through sampling using a variety of investigative processes (VCMSP298)	• investigating the uses of random sampling to collect data
Explore the variation of means and proportions of random samples drawn from the same population (VCMSP299)	• using sample properties to predict characteristics of the population
Investigate the effect of individual data values including outliers, on the range, mean and median (VCMSP300)	• using displays of data to explore and investigate effects • exploring the effect of outliers on the range for different sets of data by comparing its value with and without outliers included

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Level 8 Achievement Standard

Number and Algebra

Students use efficient mental and written strategies to make estimates and carry out the four operations with integers, and apply the index laws to whole numbers. They identify and describe rational and irrational numbers in context. Students estimate answers and solve everyday problems involving profit and loss rates, ratios and percentages, with and without the use of digital technology. They simplify a variety of algebraic expressions and connect expansion and factorisation of linear expressions. Students solve linear equations and graph linear relationships on the Cartesian plane.

Measurement and Geometry

Students convert between units of measurement for area and for volume. They find the perimeter and area of parallelograms, rhombuses and kites. Students name the features of circles, calculate circumference and area, and solve problems relating to the volume of prisms. They make sense of time duration in real applications, including the use of 24-hour time. Students identify conditions for the congruence of triangles and deduce the properties of quadrilaterals. They use tools, including digital technology, to construct congruent shapes.

Statistics and Probability

Students explain issues related to the collection of sample data and discuss the effect of outliers on means and medians of the data. They use various approaches, including the use of digital technology, to generate simple random samples from a population. Students model situations with Venn diagrams and two-way tables and explain the use of 'not', 'and' and 'or'. Students choose appropriate language to describe events and experiments. They determine complementary events and calculate the sum of probabilities.

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Level 9

In Level 9, students develop familiarity with a broader range of non-linear and linear functions and relations, and related algebra and graphs.

Students apply index laws with integer indices to a range of numerical expressions and extend this to algebraic expressions involving numbers and pro-numerals. They use indices to express very large and very small numbers in scientific notation, and apply this in measurement contexts. Students solve problems involving direct proportion and rates, and simple interest. They apply coordinate geometry to finding the distance between two points in the Cartesian plane, and the midpoint and gradient of a line segment joining two points. Students graph linear relations and solve linear equations, using tables of values, graphs and algebra. They graph simple non-linear relations such as parabolas, the reciprocal function, and circles at the origin, and solve simple related equations with and without the use of digital technology.

Students find areas of composite shapes and the surface area and volumes of right prisms and cylinders. They solve problems involving very small and very large time scales and intervals, and use scientific notation in this context. Students use similarity, enlargement transformations and apply geometric reasoning to solve problems involving ratio and scale factors. They use Pythagoras theorem and trigonometry ratios to solve problems in the plane involving right angles triangles, and develop an understanding that these involve irrational real numbers, which are generally represented by rational approximations specified to a given accuracy.

Students list outcomes for two-step experiments involving selections with and without replacement, using arrays and tree diagrams, and determine related probabilities. They use Venn diagrams and two-way tables to calculate probabilities and relative frequencies from collected or given data to estimate probabilities. They identify issues and questions involving categorical and numerical data, use back-to-back stem-plots and histograms to describe and compare the distribution of data in terms of location (centre), spread and symmetry or skew.

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Level 9 Content Descriptions

Number and Algebra

Real numbers	Elaborations
Solve problems involving direct proportion. Explore the relationship between graphs and equations corresponding to simple rate problems (VCMNA301)	• identifying direct proportion in real-life contexts
Apply index laws to numerical expressions with integer indices (VCMNA302)	• simplifying and evaluating numerical expressions, using involving both positive and negative integer indices
Express numbers in scientific notation (VCMNA303)	• representing extremely large and small numbers in scientific notation, and numbers expressed in scientific notation as whole numbers or decimals

Money and financial mathematics	Elaborations
Solve problems involving simple interest (VCMNA304)	• understanding that financial decisions can be assisted by mathematical calculations

Patterns and algebra	Elaborations
Extend and apply the index laws to variables, using positive integer indices and the zero index (VCMNA305)	• understanding that index laws apply to variables as well as numbers
Apply the distributive law to the expansion of algebraic expressions, including binomials, and collect like terms where appropriate (VCMNA306)	• understanding that the distributive law can be applied to algebraic expressions as well as numbers • understanding the relationship between expansion and factorisation and identifying algebraic factors in algebraic expressions
Apply set structures to solve real-world problems (VCMNA307)	• using a sort algorithm to determine the median of a set of numbers • exploring variation in proportion and means of random samples, drawn from a population

Linear and non-linear relationships	Elaborations
Find the distance between two points located on a Cartesian plane using a range of strategies, including graphing software (VCMNA308)	• investigating graphical and algebraic techniques for finding distance between two points • using Pythagoras' theorem to calculate distance between two points

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Find the midpoint and gradient of a line segment (interval) on the Cartesian plane using a range of strategies, including graphing software (VCMNA309)	• investigating graphical and algebraic techniques for finding midpoint and gradient • recognising that the gradient of a line is the same as the gradient of any line segment on that line
Sketch linear graphs using the coordinates of two points and solve linear equations (VCMNA310)	• determining linear rules from suitable diagrams, tables of values and graphs and describing them using both words and algebra
Graph simple non-linear relations with and without the use of digital technologies and solve simple related equations (VCMNA311)	• graphing parabolas, and circles connecting x-intercepts of a graph to a related equation

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Measurement and Geometry

Using units of measurement	Elaborations
Calculate the areas of composite shapes (VCMMG312)	• understanding that partitioning composite shapes into rectangles and triangles is a strategy for solving problems involving area
Calculate the surface area and volume of cylinders and solve related problems (VCMMG313)	• analysing nets of cylinders to establish formulas for surface area • connecting the volume and capacity of a cylinder to solve authentic problems
Solve problems involving the surface area and volume of right prisms (VCMMG314)	• solving practical problems involving surface area and volume of right prisms
Investigate very small and very large time scales and intervals (VCMMG315)	• investigating the usefulness of scientific notation in representing very large and very small numbers

Geometric reasoning	Elaborations
Use the enlargement transformation to explain similarity and develop the conditions for triangles to be similar (VCMMG316)	• establishing the conditions for similarity of two triangles and comparing this to the conditions for congruence • using the properties of similarity and ratio, and correct mathematical notation and language, to solve problems involving enlargement. For example, scale diagrams • using the enlargement transformation to establish similarityunderstanding that similarity and congruence help describe relationships between geometrical shapes and are important elements of reasoning and proof
Solve problems using ratio and scale factors in similar figures (VCMMG317)	• establishing the relationship between areas of similar figures and the ratio of corresponding sides (scale factor)

Pythagoras and trigonometry	Elaborations
Investigate Pythagoras' Theorem and its application to solving simple problems involving right angled triangles (VCMMG318)	• understanding that Pythagoras' Theorem is a useful tool in determining unknown lengths in right-angled triangles and has widespread applications • recognising that right-angled triangle calculations may generate results that can be integers, fractions or irrational numbers
Use similarity to investigate the constancy of the sine, cosine and tangent ratios for a given angle in right-angled triangles (VCMMG319)	• developing understanding of the relationship between the corresponding sides of similar right-angled triangles
Apply trigonometry to solve right-angled triangle problems (VCMMG320)	• understanding the terms 'adjacent' and 'opposite' sides in a right-angled triangle • selecting and accurately using the correct trigonometric ratio to find unknown sides (adjacent, opposite and hypotenuse) and angles in right-angled triangles

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Statistics and Probability

Chance	Elaborations
List all outcomes for two-step chance experiments, both with and without replacement using tree diagrams or arrays. Assign probabilities to outcomes and determine probabilities for events (VCMSP321)	• conducting two-step chance experiments • using systematic methods to list outcomes of experiments and to list outcomes favourable to an event • comparing experiments which differ only by being undertaken with replacement or without replacement
Calculate relative frequencies from given or collected data to estimate probabilities of events involving 'and' or 'or' (VCMSP322)	• using Venn diagrams or two-way tables to calculate relative frequencies of events involving ‘and’, ‘or’ questions • using relative frequencies to find an estimate of probabilities of ‘and’, ‘or’ events
Investigate reports of surveys in digital media and elsewhere for information on how data were obtained to estimate population means and medians (VCMSP323)	• investigating a range of data and its sources. For example, the age of residents in Australia, Cambodia and Tonga, or the number of subjects studied at school by 14-year-old students in Australia, Japan and Timor-Leste

Data representation and interpretation	Elaborations
Identify everyday questions and issues involving at least one numerical and at least one categorical variable, and collect data directly from secondary sources (VCMSP324)	• comparing the annual rainfall in various parts of Australia, Pakistan, New Guinea and Malaysia
Construct back-to-back stem-and-leaf plots and histograms and describe data, using terms including ‘skewed’, ‘symmetric’ and ‘bi modal’ (VCMSP325)	• using stem-and-leaf plots to compare two like sets of data such as the heights of girls and the heights of boys in a class • describing the shape of the distribution of data using terms such as ‘positive skew’, ‘negative skew’ and 'symmetric' and 'bi-modal'
Compare data displays using mean, median and range to describe and interpret numerical data sets in terms of location (centre) and spread (VCMSP326)	• comparing means, medians and ranges of two sets of numerical data which have been displayed using histograms, dot plots, or stem and leaf plots

Level 9 Achievement Standard

Number and Algebra

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Students apply the index laws using integer indices to variables and numbers, express numbers in scientific notation, solve problems involving very small and very large numbers, and check the order of magnitude of calculations. They solve problems involving simple interest. Students use the distributive law to expand algebraic expressions, including binomial expressions, and simplify a range of algebraic expressions. They find the distance between two points on the Cartesian plane and the gradient and midpoint of a line segment using a range of strategies including the use of digital technology. Students sketch and draw linear and non-linear relations, solve simple related equations and explain the relationship between the graphical and symbolic forms, with and without the use of digital technology.

Measurement and Geometry

Students solve measurement problems involving perimeter and area of composite shapes, surface area and volume of rectangular prisms and cylinders, with and without the use of digital technology. They relate three-dimensional objects to two-dimensional representations. Students explain similarity of triangles, interpret ratios and scale factors in similar figures, and apply Pythagoras's theorem and trigonometry to solve problems involving angles and lengths in right-angled triangles.

Statistics and Probability

Students compare techniques for collecting data from primary and secondary sources, and identify questions and issues involving different data types. They construct histograms and back-to-back stem-and-leaf plots with and without the use of digital technology. Students identify mean and median in skewed, symmetric and bi-modal displays and use these to describe and interpret the distribution of the data. They calculate relative frequencies to estimate probabilities. Students list outcomes for two-step experiments and assign probabilities for those outcomes and related events.

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Level 10

In Level 10, students extend their use of mathematical models to a wide range of familiar and unfamiliar contexts, involving the use of all types of real numbers. They recognise the role of logical argument and proof in establishing mathematical propositions. Students apply mental, written or technology-assisted forms of computation as appropriate, and routinely use estimation to validate or provide bounds for their answers. They use exponential functions to model compound interest problems.

Students expand, factorise, simplify and substitute into a wide range of algebraic expressions, including linear, quadratic, and exponential terms and relations, as well as simple algebraic fractions with numerical denominators. They solve related equations, linear inequalities and simultaneous linear equations, with and without the use of digital technology. They explore the connection between tabular, graphical and algebraic representations of non-linear relations, including circles with centres at any location in the Cartesian plane.

Students solve problems involving surface area and volume for a range of objects, and follow proofs of key geometric results involving the application of congruence and similarity. They solve practical problems in two and three dimensions involving right angles triangles, Pythagoras theorem and trigonometry.

Students extend their work in probability to combinations of up to three events, using lists, tables, Venn diagrams, tree diagrams and grids as applicable to determine probabilities. They explore the concepts of conditional probability and independence, and their application to solving problems involving chance events.

Students use quartiles and the interquartile range as a measure of spread, and construct and interpret boxplots to compare data sets. They relate box plots to corresponding dot plots and histograms. Students explore the association between two numerical variables using scatterplots, in particular with time as the independent variable. They discuss claims made using statistics in various media articles and other reports, on issues of interest.

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Level 10 Content Descriptions

Number and Algebra

Real numbers	Elaborations
Solve simple problems involving inverse proportion (VCMNA327)	• identifying inverse proportion in real life contexts such as exchange rates • modelling problems involving inverse proportion and solving related equations

Money and financial mathematics	Elaborations
Connect the compound interest formula to repeated applications of simple interest using appropriate digital technologies (VCMNA328)	• working with authentic information, data and interest rates to calculate compound interest and solve related problems

Patterns and algebra	Elaborations
Factorise algebraic expressions by taking out a common algebraic factor (VCMNA329)	• using the distributive law and the index laws to factorise algebraic expressions • understanding the relationship between factorisation and expansion
Simplify algebraic products and quotients using index laws (VCMNA330)	• applying knowledge of index laws to algebraic terms, and simplifying algebraic expressions using both positive and negative integral indices
Apply the four operations to simple algebraic fractions with numerical denominators (VCMNA331)	• expressing the sum and difference of algebraic fractions with a common denominator • using the index laws to simplify products and quotients of algebraic fractions
Expand binomial products and factorise monic quadratic expressions using a variety of strategies (VCMNA332)	• exploring the method of completing the square to factorise quadratic expressions and solve quadratic equations • identifying and using common factors, including binomial expressions, to factorise algebraic expressions using the technique of grouping in pairs • using the identities for perfect squares and the difference of squares to factorise quadratic expressions
Substitute values into formulas to determine an unknown and re-arrange formulas to solve for a particular term (VCMNA333)	• solving simple equations arising from formulas • re-arranging expressions to make a specified variable the subject such as calculating the radius of a sphere to produce a given volume
Implement algorithms using data structures in a general-purpose programming language (VCMNA334)	• using two-dimensional arrays such as matrices to represent and implement sequences of transformations of sets of points in the plane • using pointers in algorithms

Linear and non-linear relationships	Elaborations

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Solve problems involving linear equations, including those derived from formulas (VCMNA335)	• representing word problems with simple linear equations and solving them to answer questions
Solve linear inequalities and graph their solutions on a number line (VCMNA336)	• representing word problems with simple linear inequalities and solving them to answer questions
Solve simultaneous linear equations, using algebraic and graphical techniques including using digital technology (VCMNA337)	• associating the solution of simultaneous equations with the coordinates of the intersection of their corresponding graphs
Solve problems involving gradients of parallel and perpendicular lines (VCMNA338)	• solving problems using the fact that parallel lines have the same gradient and conversely that if two lines have the same gradient then they are parallel • solving problems using the fact that the product of the gradients of perpendicular lines is –1 and conversely that if the product of the gradients of two lines is –1 then they are perpendicular
Explore the connection between algebraic and graphical representations of relations such as simple quadratic, reciprocal, circle and exponential, using digital technology as appropriate (VCMNA339)	• sketching graphs of parabolas, and circles • applying translations, reflections and stretches to parabolas and circles • sketching the graphs of exponential functions using transformations • plotting graphs of families of relations where the product of two variable is equal to a fixed constant
Solve linear equations involving simple algebraic fractions (VCMNA340)	• solving a wide range of linear equations, including those involving one or two simple algebraic fractions, and checking solutions by substitution • representing word problems, including those involving fractions, as equations and solving them to answer the question
Solve simple quadratic equations using a range of strategies (VCMNA341)	• using a variety of techniques to solve quadratic equations, including grouping, completing the square, the quadratic formula, and choosing two integers with the required product and sum
Solve equations using systematic guess-check-and-refine with digital technology (VCMNA342)	• refining intervals on graphs and/or in tables of values to determine with increasing accuracy when the values of two functions are approximately equal

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Measurement and Geometry

Using units of measurement	Elaborations
Solve problems involving surface area and volume for a range of prisms, cylinders and composite solids (VCMMG343)	• investigating and determining the volumes and surface areas of composite solids by considering the individual solids from which they are constructed

Geometric reasoning	Elaborations
Formulate proofs involving congruent triangles and angle properties (VCMMG344)	• applying an understanding of relationships to deduce properties of geometric figures (for example the base angles of an isosceles triangle are equal)
Apply logical reasoning, including the use of congruence and similarity, to proofs and numerical exercises involving plane shapes (VCMMG345)	• distinguishing between a practical demonstration and a proof (for example demonstrating triangles are congruent by placing them on top of each other, as compared to using congruence tests to establish that triangles are congruent) • performing a sequence of steps to determine an unknown angle giving a justification in moving from one step to the next. • communicating a proof using a sequence of logically connected statements

Pythagoras and trigonometry	Elaborations
Solve right-angled triangle problems including those involving direction and angles of elevation and depression (VCMMG346)	• applying Pythagoras's Theorem and trigonometry to problems in surveying and design

Statistics and Probability

Chance	Elaborations
Describe the results of two- and three-step chance experiments, both with and without replacements, assign probabilities to outcomes and determine probabilities of events. Investigate the concept of independence (VCMSP347)	• recognising that an event can be dependent on another event and that this will affect the way its probability is calculated
Use the language of ‘if ....then, ‘given’, ‘of’, ‘knowing that’ to investigate conditional statements and identify common mistakes in interpreting such language (VCMSP348)	• using two-way tables and Venn diagrams to understand conditional statements • using arrays and tree diagrams to determine probabilities

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Data representation and interpretation	Elaborations
Determine quartiles and interquartile range and investigate the effect of individual data values, including outliers on the interquartile range (VCMSP349)	• finding the five-number summary (minimum and maximum values, median and upper and lower quartiles) and using its graphical representation, the box plot, as tools for both numerically and visually comparing the centre and spread of data sets • exploring the effect of varying data values, including outliers, on the interquartile range for different sets of data
Construct and interpret box plots and use them to compare data sets (VCMSP350)	• understanding that box plots are an efficient and common way of representing and summarising data and can facilitate comparisons between data sets • using parallel box plots to compare data about the age distribution of Aboriginal and Torres Strait Islander people with that of the Australian population as a whole
Compare shapes of box plots to corresponding histograms and dot plots and discuss the distribution of data (VCMSP351)	• investigating data in different ways to make comparisons and draw conclusions • using a dot plot, box-plot or histogram to construct a cumulative frequency distribution for a set of data
Use scatter plots to investigate and comment on relationships between two numerical variables (VCMSP352)	• using authentic data to construct scatter plots, make comparisons and draw conclusions
Investigate and describe bivariate numerical data, including where the independent variable is time (VCMSP353)	• investigating biodiversity changes in Australia since European occupation • constructing and interpreting data displays representing bivariate data over time • constructing scatter-plots for two numerical variables and investigate trends such as water storage levels over time or weight and height distributions
Evaluate statistical reports in the media and other places by linking claims to displays, statistics and representative data (VCMSP354)	• investigating the use of statistics in reports regarding the growth of Australia's trade with other countries of the Asia region • evaluating statistical reports comparing the life expectancy of Aboriginal and Torres Strait Islander people with that of the Australian population as a whole

Level 10 Achievement Standard

Number and Algebra

Students recognise the connection between simple and compound interest. They solve problems involving linear equations and inequalities, quadratic equations and pairs of simultaneous linear equations and related graphs, with and without the use of digital technology. Students substitute into formulas, find unknown values, manipulate linear algebraic expressions, expand binomial expressions and factorise monic and simple non-monic quadratic expressions, with and without the use of digital technology. They represent linear, quadratic and exponential functions numerically, graphically and algebraically, and use them to model situations and solve practical problems.

Measurement and Geometry

Students solve and explain surface area and volume problems relating to composite solids. They use parallel and perpendicular lines, angle and triangle properties, similarity, trigonometry and congruence to solve practical problems and develop proofs involving lengths, angles and areas in plane shapes. They use digital technology to construct and manipulate geometric shapes and objects, and explore symmetry and pattern in two dimensions.

Statistics and Probability

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Students compare univariate data sets by referring to summary statistics and the shape of their displays. They describe bivariate data where the independent variable is time and use scatter-plots generated by digital technology to investigate relationships between two continuous variables. Students evaluate the use of statistics in the media. They list outcomes for multi-step chance experiments involving independent and dependent events, and assign probabilities for these experiments.

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Level 10A

Level 10A provides optional, additional content for students to be extended in their mathematical studies.

Students could extend work in number and algebra to investigate the structure and properties of number systems, with further analysis of order relations and inequalities. They could extend the study of trigonometry to include an introduction to circular functions and equations, or extend the study of indices and exponential functions to logarithms, including an introduction to logarithmic functions.

Students could extend work in measurement and geometry to proving a broader range of geometric propositions solving trigonometric problems in non-right angles triangles, or solving three dimensional problems involving surface area and volume of cones and spheres and composite shapes.

Students could extend work in statistics and probability to explore the concepts of conditionality, dependence and independence in depth, or consider how various measures of location and spread can be used to describe the distribution of a data set, and investigate how robust these are with respect to variation in the data, in particular with respect to measurement error.

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Level 10A Content Descriptions

Number and Algebra

Real numbers	Elaborations
Define rational and irrational numbers and perform operations with surds and fractional indices (VCMNA355)	• understanding that the real number system includes irrational numbers • extending the index laws to rational number indices • performing the four operations with surds
Use the definition of a logarithm to establish and apply the laws of logarithms and investigate logarithmic scales in measurement (VCMNA356)	• investigating the relationship between exponential and logarithmic expressions • simplifying expressions using the logarithm laws • investigating the use of logarithmic scales to represent very small and very large quantities

Patterns and algebra	Elaborations
Investigate the concept of a polynomial and apply the factor and remainder theorems to solve problems (VCMNA357)	• investigating the relationship between algebraic long division and the factor and remainder theorems
Devise and use algorithms and simulations to solve mathematical problems (VCMNA358)	• applying a systematic guess-check-and-refine algorithm to identify an approximate value for the root of an equation in an interval • developing simulations for counter-intuitive problems in probability such as the Monty Hall problem or derangements

Linear and non-linear relationships	Elaborations
Describe, interpret and sketch parabolas, hyperbolas, circles and exponential functions and their transformations (VCMNA359)	• applying transformations, including translations, reflections in the axes and stretches to help graph parabolas, rectangular hyperbolas, circles and exponential functions
Solve simple exponential equations (VCMNA360)	• investigating exponential equations derived from authentic mathematical models based on population growth
Apply understanding of polynomials to sketch a range of curves and describe the features of these curves from their equation (VCMNA361)	• investigating the features of graphs of polynomials including axes intercepts and the effect of repeated factors
Factorise monic and non-monic quadratic expressions and solve a wide range of quadratic equations derived from a variety of contexts (VCMNA362)	• writing quadratic equations that represent practical problems

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Use function notation to describe the relationship between dependent and independent variables in modelling contexts (VCMNA363)	• identify independent and dependent variables in modelling contexts and represent the relation between them using tables, graphs and rules • using technology to draw graphs of functions defined using function notation with consideration of domain and range
Solve simultaneous equations using systematic guess-check-and-refine with digital technology (VCMNA364)	• using graphs to determine a convergent set of intervals which contain a point of intersection of the graphs of two functions • using cobweb diagram to solve simultaneous equations numerically

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Measurement and Geometry

Using units of measurement	Elaborations
Solve problems involving surface area and volume of right pyramids, right cones, spheres and related composite solids (VCMMG365)	• using formulas to solve problems • using authentic situations to apply knowledge and understanding of surface area and volume

Geometric reasoning	Elaborations
Prove and apply angle and chord properties of circles (VCMMG366)	• performing a sequence of steps to determine an unknown angle or length in a diagram involving a circle, or circles, giving a justification in moving from one step to the next • communicating a proof using a logical sequence of statements • proving results involving chords of circles

Pythagoras and trigonometry	Elaborations
Establish the sine, cosine and area rules for any triangle and solve related problems (VCMMG367)	• applying knowledge of sine, cosine and area rules to authentic problems such as those involving surveying and design
Use the unit circle to define trigonometric functions as functions of a real variable, and graph them with and without the use of digital technologies (VCMMG368)	• establishing the symmetrical properties of trigonometric functions • investigating angles of any magnitude • understanding that trigonometric functions are periodic and that this can be used to describe motion • identifying points on the unit circle via arc lengths in radians, which correspond to specified values of the circular functions sine, cosine and tangent
Solve simple trigonometric equations (VCMMG369)	• using periodicity and symmetry to solve equations
Apply Pythagoras' theorem and trigonometry to solving three-dimensional problems in right-angled triangles (VCMMG370)	• investigating the applications of Pythagoras's theorem in authentic problems

Statistics and Probability

Chance	Elaborations
Investigate reports of studies in digital media and elsewhere for information on their planning and implementation (VCMSP371)	• evaluating the appropriateness of sampling methods in reports where statements about a population are based on a sample • evaluating whether graphs in a report could mislead, and whether graphs and numerical information support the claims

Data representation and interpretation	Elaborations

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Calculate and interpret the mean and standard deviation of data and use these to compare data sets. Investigate the effect of individual data values including outliers, on the standard deviation (VCMSP372)	• using the standard deviation to describe the spread of a set of data • using the mean and standard deviation to compare numerical data sets • constructing distributions for the mean and standard deviation of simple random samples from a population
Use digital technology to investigate bivariate numerical data sets. Where appropriate use a straight line to describe the relationship allowing for variation, make predictions based on this straight line and discuss limitations (VCMSP373)	• investigating different techniques for finding a ‘line of best fit’ • using a fitted line to data to make predictions between and beyond existing data value sand discuss limitations of these predictions

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